

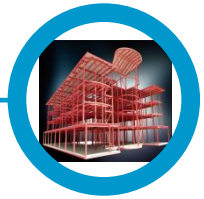


STEEL STRUCTURES/ SPACE STRUCTURES (CE-862) Fall Semester

Lecture 03



Axially Loaded Compression Members



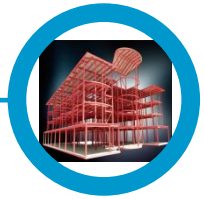
What is Compression Member?

- Compression members are found in bridges, roof trusses, towers and bracing system
- Design for buckling
- Design Strength in Tension member
    a) Yielding in gross section
    b) Fracture in net section





Compression Members



This end view of the Sainsbury Centre for Visual Arts, Norwich, England clearly shows its extruded profile and the modular cladding system. The end wall uses structural mullionless glazing. Although the concept behind the extruded form was to allow for an addition, in fact the bollards are there to prevent traffic over the lawn, as the addition that was constructed in 2000 sits below the grass lawn.

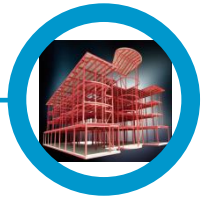


Compression Members



The structure of the Richmond Speed Skating Oval, Richmond, BC, Canada, was clearly a marriage between steel and timber.

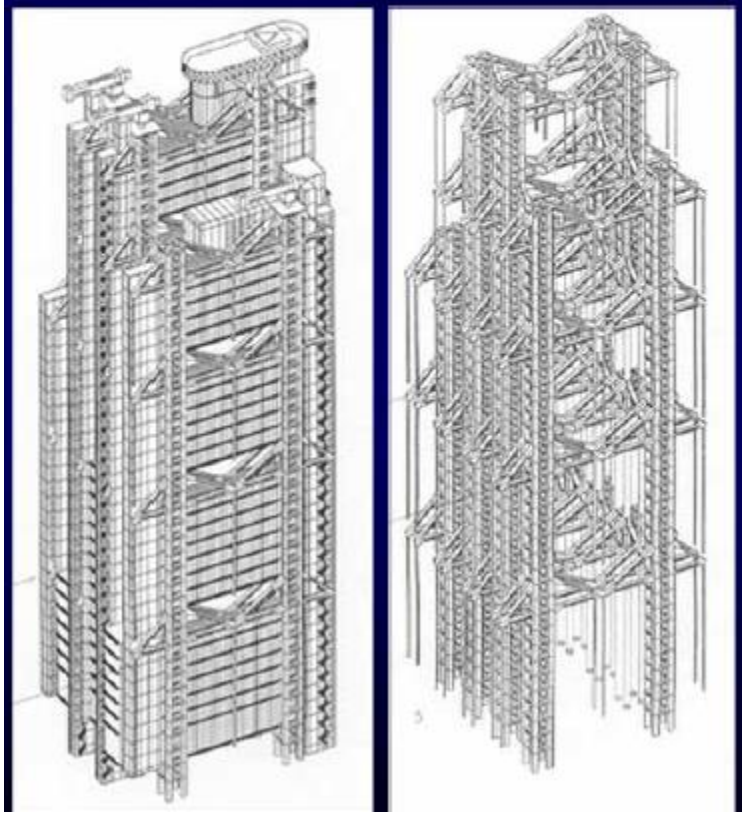
Compression Members

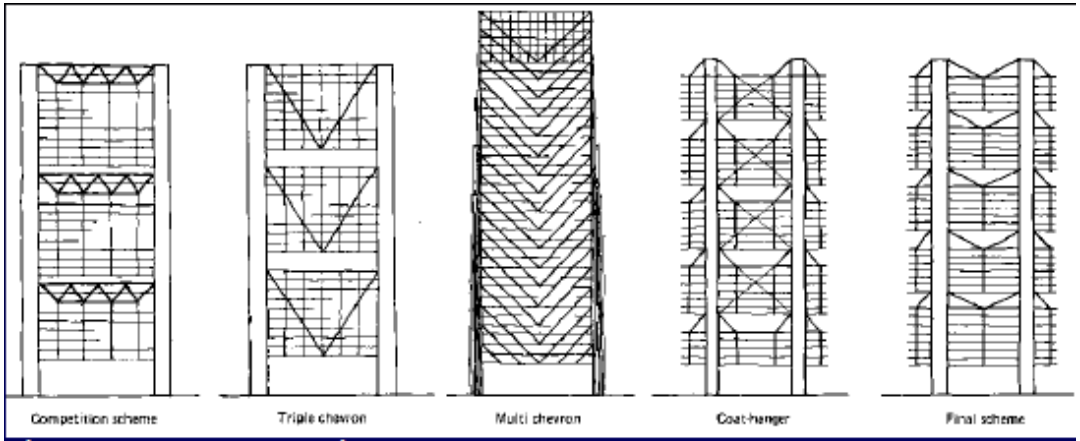


Hong Kong Shanghai Bank

Architect: Norman Foster

Engineer: Ove Arup





↑ Scheme development ↑

↓ Computer analysis ↓

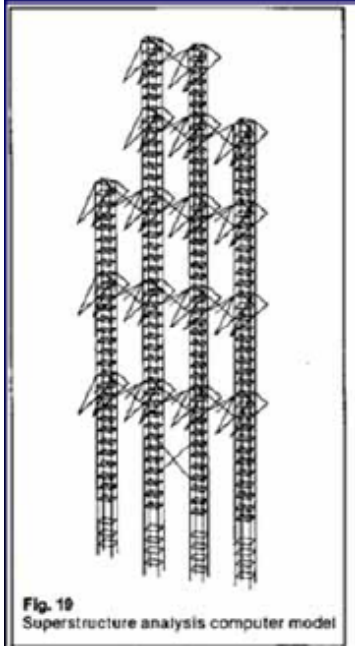


Fig. 19
Superstructure analysis computer model

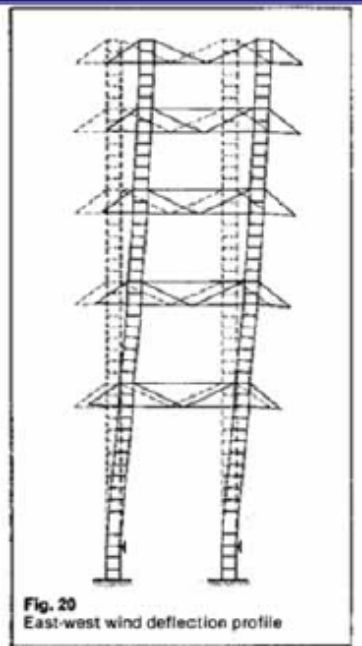


Fig. 20
East-west wind deflection profile

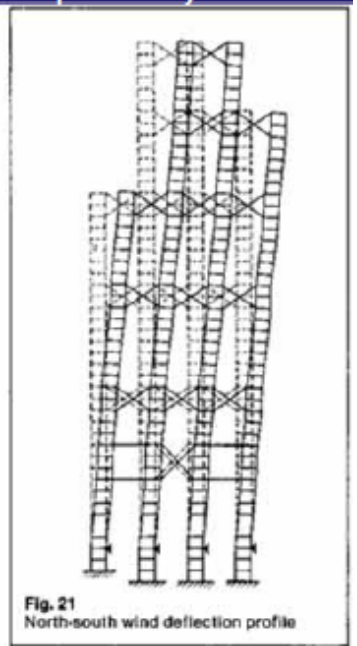
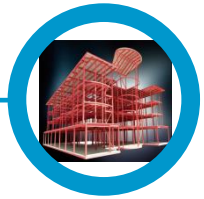


Fig. 21
North-south wind deflection profile



Types of Compression Member

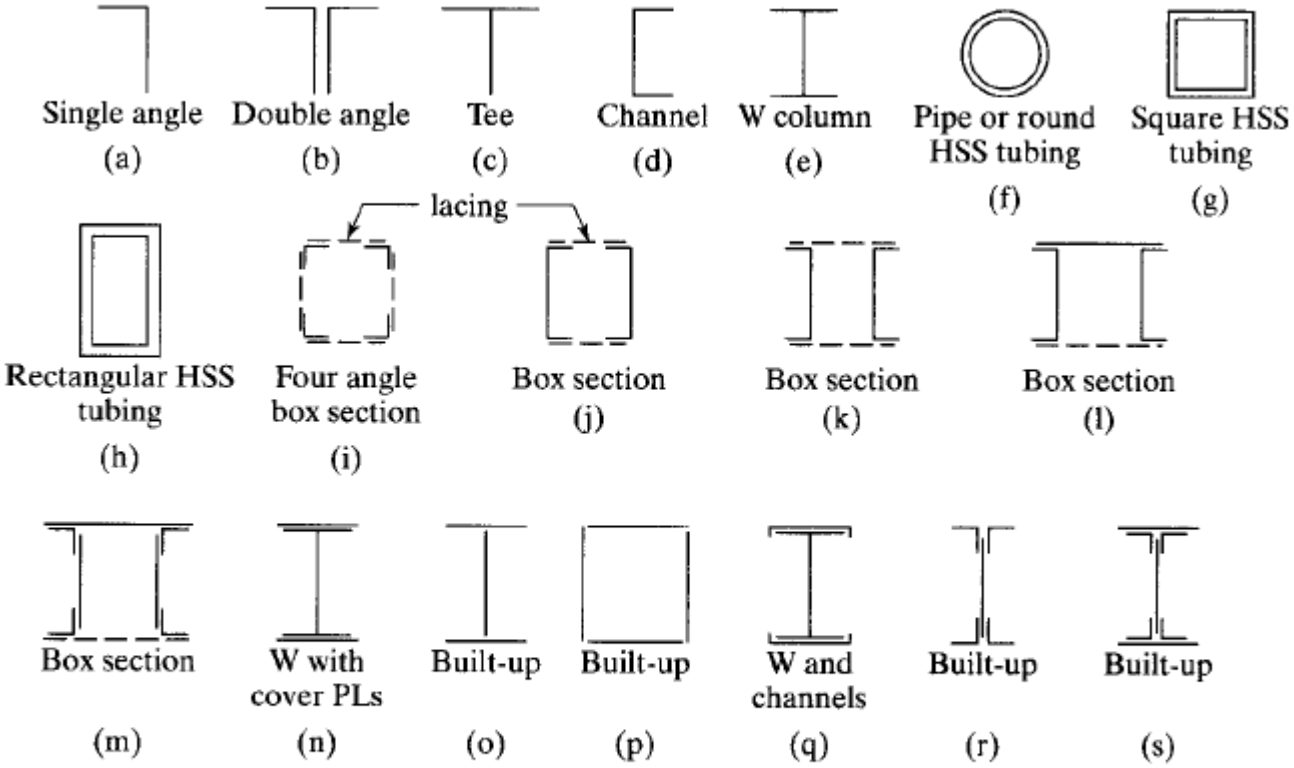
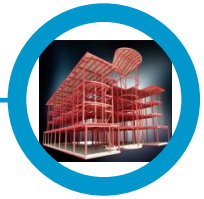


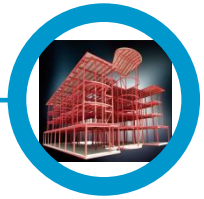
FIGURE 5.2

Types of compression members.



Types of Compression Member

1. The most efficient compression member is one that has a constant radius of gyration about its centroid, a property available in round HSS tubing and pipe sections. Square tubing is the next-most-efficient compression member.
2. Four-sided and round sections are much easier to paint than are the six-sided open W, S, and M sections. Furthermore, the rounded corners make it easier to apply paint or other coatings uniformly around the sections.
3. They have less surface area to paint or fireproof.
4. They have excellent torsional resistance.
5. The surfaces of tubing are quite attractive.
6. When exposed, the round sections have wind resistance of only about two-thirds of that of flat surfaces of the same width.
7. If cleanliness is important, hollow structural tubing is ideal, as it doesn't have the problem of dirt collecting between the flanges of open structural shapes.

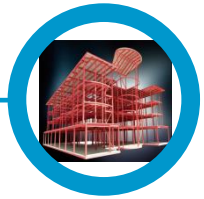


Types of Compression Member

- As we'll discover throughout this series of lectures, column strength is direct function of its length.
  - More specifically, its slenderness.

- Only columns that are "short" can be loaded to the yield stress.
- "Longer" columns will usually buckle before they yield.
  - "Medium length" columns have inelastic buckling capacity.
  - "Slender" columns buckle elastically.

- To begin, we must first discuss the concept of buckling.

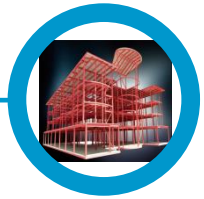


Types of Compression Member

- A compression member that is so slender compared to its length that under gradually increasing loads it fails by buckling at loads considerably less than those required for crushing.

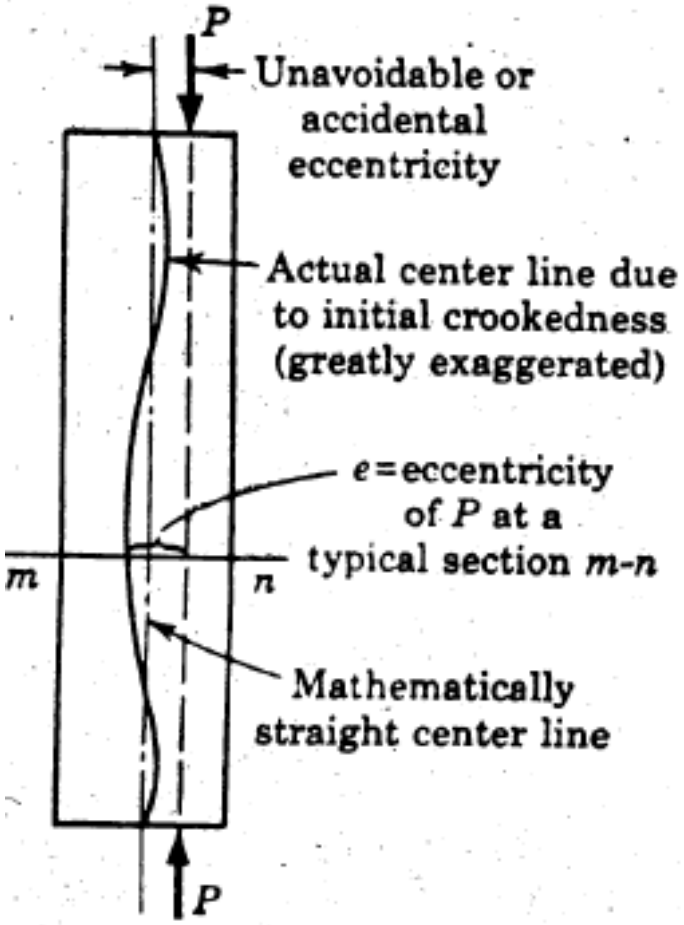
- Buckling is a sudden loss of stiffness due to compressive loads.
  - Leads into a major topic in structural engineering: stability.
  - Can lead to sudden failures of structures.

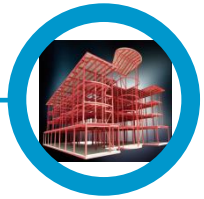
- Critical buckling loads were first derived by Leonard Euler in 1749.



Development of column formulas

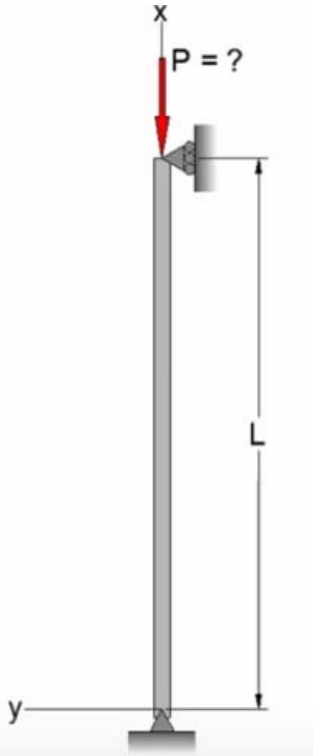
- Columns are divided into three groups: long, intermediate and short
- Ideal column is homogeneous, constant cross-section and subject to axial compressive load.
- Actual column has small initial imperfections

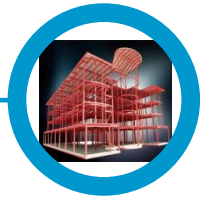




Development of column formulas

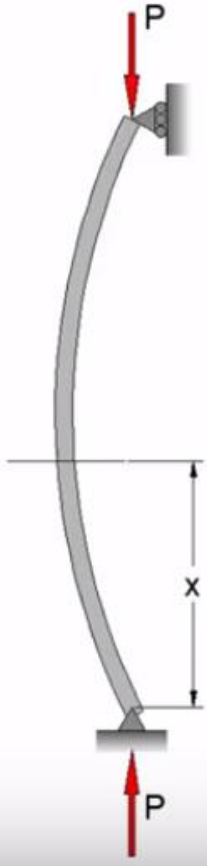
- Let's consider a simply-supported column subjected to an axial load "P."
  - EI = constant
- The goal is to determine the load "P" and its associated stress that causes buckling.

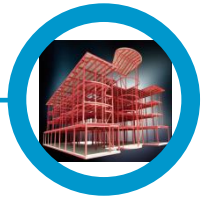




Development of column formulas

- Let's cut a section at a distance "x" from the bottom end of the column (in the column's buckled state).
  - It's a fair assumption that the column would look something like this:
- Note that the vertical reaction will also be "P" from statics.





Development of column formulas

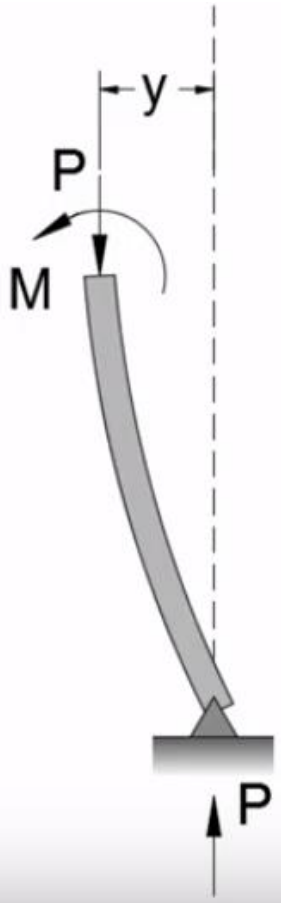
- If we cut a section and sum moments:
  - Note the internal moment drawn in the positive direction:

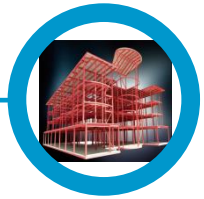
$$\sum M_{cut} = 0$$

↻	↺
0	M
	Py

$$M + Py = 0$$

$$\therefore \boxed{M = -Py}$$





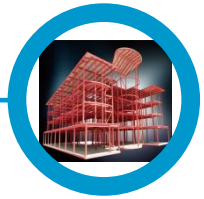
Development of column formulas

- Also, recall our moment curvature relationship from beam deflection:

$$\frac{d^2y}{dx^2} = \frac{M}{EI} \qquad \text{or} \qquad y'' = \frac{M}{EI}$$

  - If our expression for moment is: $M = -Py$

  - Substituting: $y'' = \frac{-Py}{EI}$



Development of column formulas

- Rearranging:

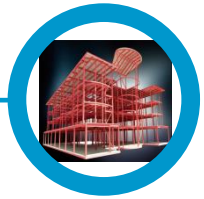
$$y'' = \frac{-Py}{EI} \qquad \text{or} \qquad y'' + \frac{P}{EI}y = 0$$

- Note:
  - To make the math simpler, we'll define an arbitrary constant:

$$\alpha^2 = \frac{P}{EI}$$

- Substituting:

$$\boxed{y'' + \alpha^2 y = 0} \leftarrow 2^{\text{nd}} \text{ Order Differential Equation}$$



Development of column formulas

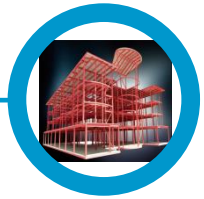
- Solving our differential equation:

$$y'' + \alpha^2 y = 0$$

$$\therefore y(x) = C_1 \cos(\alpha x) + C_2 \sin(\alpha x)$$

  - Much like beam deflections, we have two arbitrary constants.
  - Therefore, we'll need boundary conditions:
    - Since this is a simply-supported column.

$$y(0) = 0 \qquad \text{and} \qquad y(L) = 0$$



Development of column formulas

- Utilizing our boundary conditions:

$$y(x) = C_1 \cos(\alpha x) + C_2 \sin(\alpha x)$$

  - First, the bottom end of the column:

$$y(0) = C_1 \cos(0) + C_2 \sin(0)$$

$$0 = C_1 (1) + C_2 (0)$$

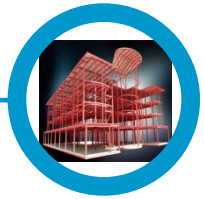
$$\therefore \boxed{C_1 = 0}$$

  - Now, the top end:

$$y(x) = C_2 \sin(\alpha x)$$

$$y(L) = C_2 \sin(\alpha L)$$

$$\boxed{C_2 \sin(\alpha L) = 0}$$

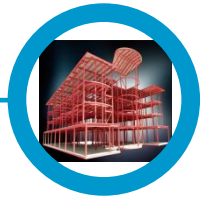


Development of column formulas

- This yields several trivial solutions to our differential equation:

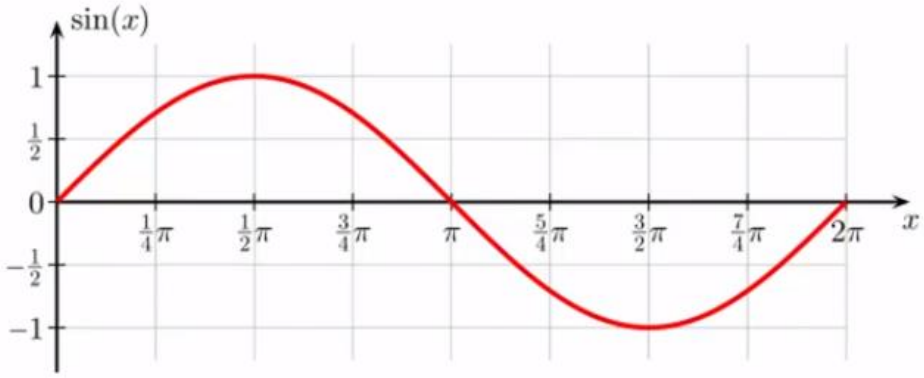
$$C_2 \sin(\alpha L) = 0$$

1. $C_2 = 0$
   - Indicates no deflection
2. $L = 0$
   - Column has no length
3. $\alpha = 0$
   - Column has no applied load
4. $\sin(\alpha L) = 0$
   - Maybe not so trivial…



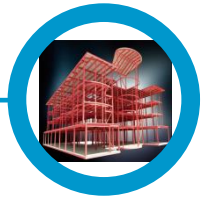
Development of column formulas

- If we recall from trigonometry:



– Therefore, if sin(αL) = 0:

$$\alpha L = n\pi \quad (n = 1,2,3\ldots) \qquad \text{or} \qquad \alpha = \frac{n\pi}{L}$$



Development of column formulas

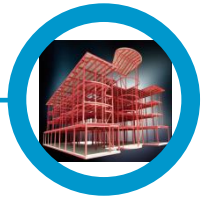
- Plugging back into our expression for “α” and solving for “P”:

$$\alpha^2 = \frac{P}{EI}$$

$$\left(\frac{n\pi}{L}\right)^2 = \frac{P}{EI}$$

$$\frac{n^2\pi^2}{L^2} = \frac{P}{EI}$$

$$\therefore P = \frac{n^2\pi^2 EI}{L^2}$$



Development of column formulas

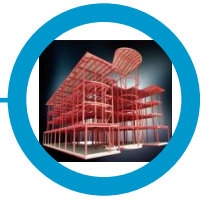
- We now have expressions for the critical load and the deflected shape:

$$y(x) = C_2 \sin(\alpha x) \qquad P = \frac{n^2 \pi^2 EI}{L^2}$$

  - Substituting the previously derived "α":

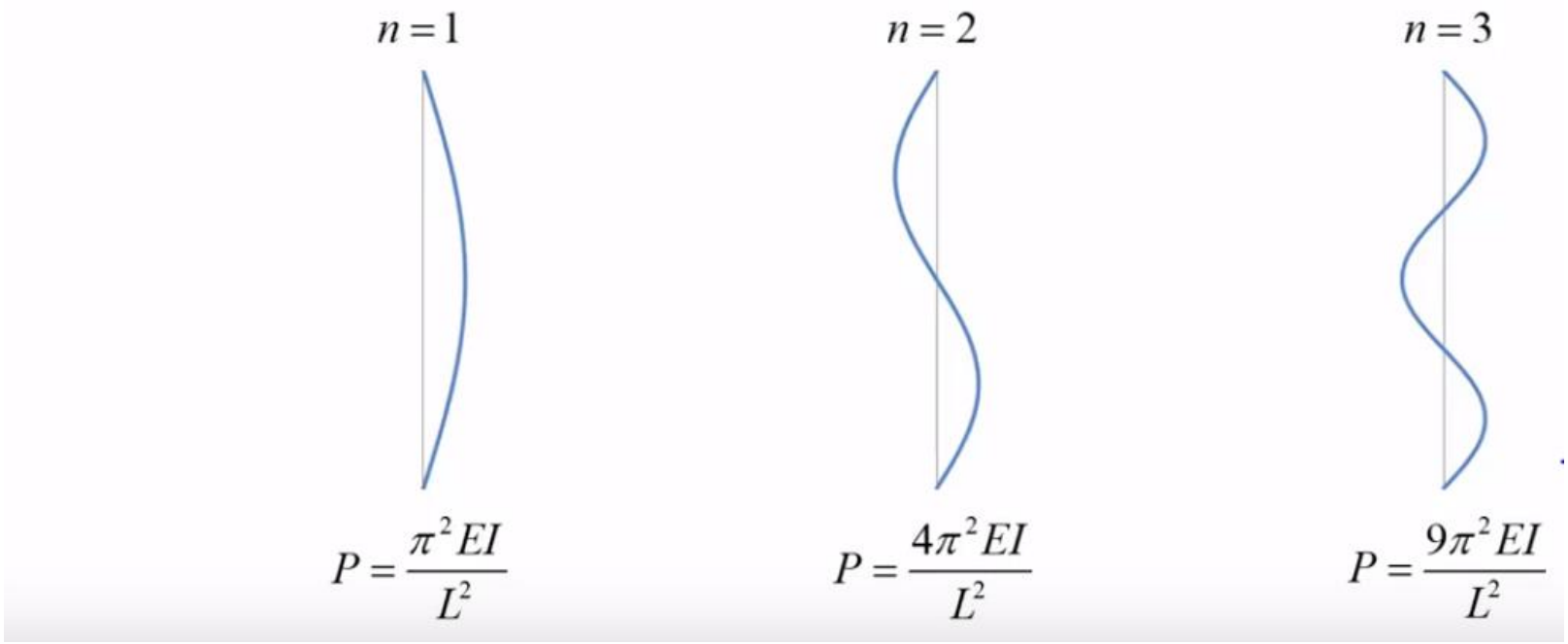
$$\therefore y(x) = C_2 \sin\left(\frac{n\pi}{L} x\right)$$

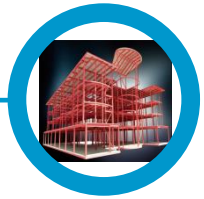
- We can now determine critical loads for different buckled shapes:
  - Also called buckling modes (i.e. one for each "n" value).



Development of column formulas

- Buckled shapes (buckling modes):





Development of column formulas

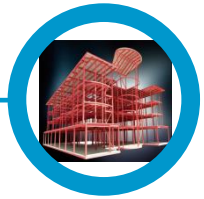
- Therefore, the buckling load for a simply-supported column is as follows:

$$P_{cr} = \frac{\pi^2 EI}{L^2}$$

  - We can also compute an equivalent buckling stress:

$$F_{cr} = \frac{P_{cr}}{A} = \frac{\pi^2 E}{L^2}\left(\frac{I}{A}\right)$$

  - This term in the parentheses is another section property:
    - Function of moment of inertia and area!



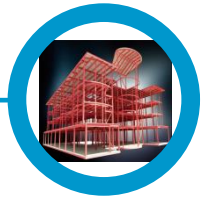
Development of column formulas

- In mechanics, recall that we define this value as a radius of gyration:
  - Specifically: $r^2 = \frac{I}{A}$ or $r = \sqrt{\frac{I}{A}}$
- Therefore, we can rewrite our buckling stress equation as follows:

$$F_{cr} = \frac{\pi^2 E}{L^2}\left(\frac{I}{A}\right) = \pi^2 E\left(\frac{r^2}{L^2}\right) = \frac{\pi^2 E}{\left(L^2/r^2\right)}$$

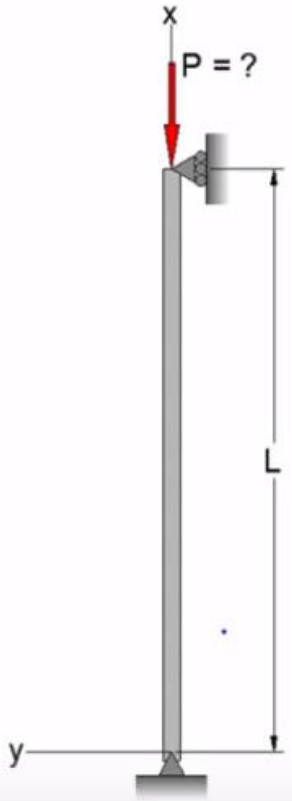
  - Note that we define "L/r" as slenderness.

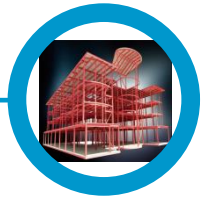
$$\therefore F_{cr} = \frac{\pi^2 E}{(L/r)^2}$$



Development of column formulas

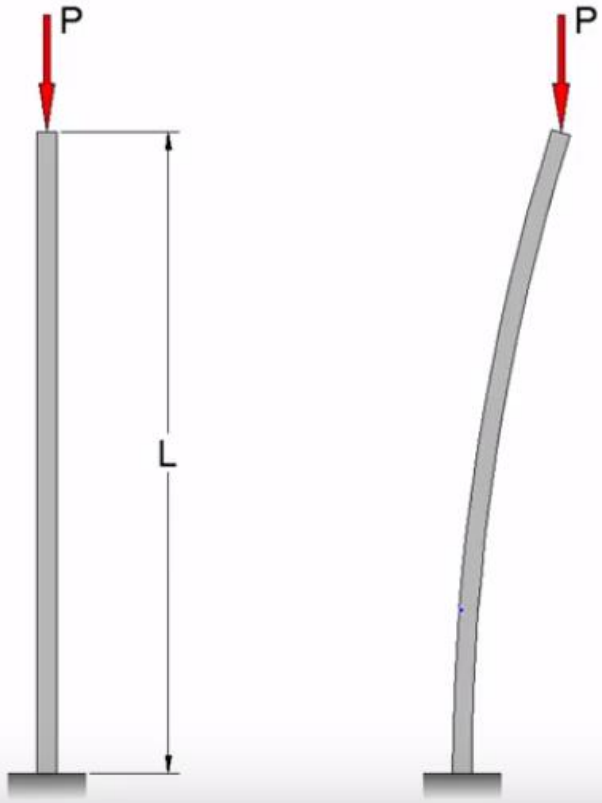
- Keep in mind, when we derived this formula, we kept this column's boundary conditions as pinned-pinned.
  - What if the boundary conditions were changed?
    - Fixed-fixed?
    - Fixed-free?
    - Etc.

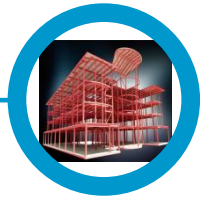




Development of column formulas

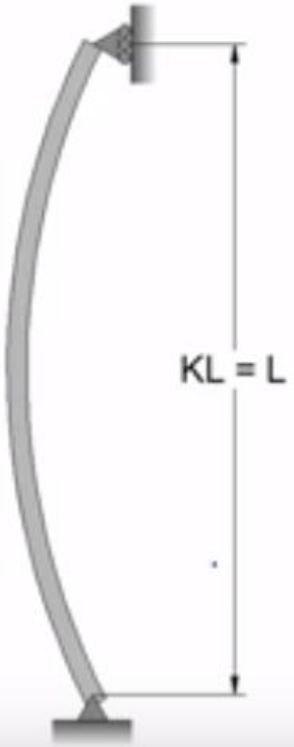
- Consider a fixed-free column:
  - It stands to reason that it's buckled shape would be as shown:
- We can relate this buckled shape to that of a pinned-pinned column.
  - This is done through the use of an effective length factor, "K".

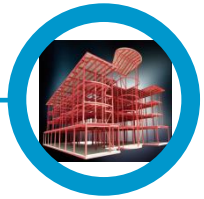




Development of column formulas

- "K" modifies the length of a column to relate it to a pinned-pinned column.
  - Specifically, this refers to the distance between inflection points.
  - For a pinned-pinned column, K = 1.0.
  - That is, the effective length is the same as the actual length.

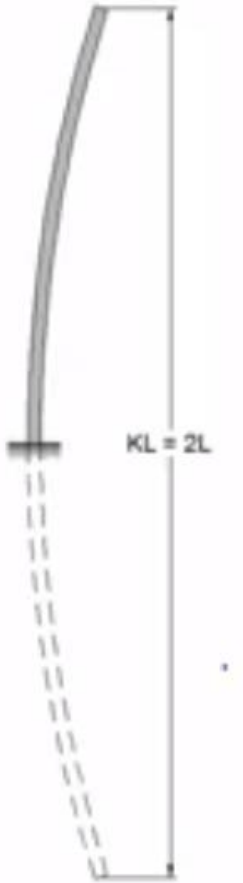


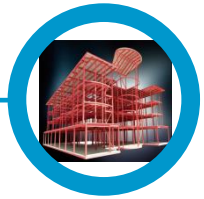


Development of column formulas

- For a fixed-free column, K = 2.0
  - That is, the effective length is the same as twice the actual length.
- Therefore, we can redefine our expressions as follows:

$$P_{cr} = \frac{\pi^2 EI}{(KL)^2} \qquad F_{cr} = \frac{\pi^2 E}{(KL/r)^2}$$





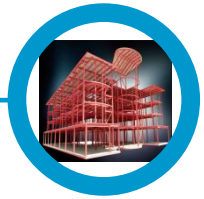
Development of column formulas

Pg. 16.1-511

- Effective length factors can be found in the commentary of the Specification.
  - Note: Always use recommended values instead of theoretical values!
- Also, keep in mind that these values are for single columns!
  - Later on, we'll discuss how to compute "K" for columns in frames!

TABLE C-A-7.1
Approximate Values of Effective Length Factor, *K*

Buckled shape of column is shown by dashed line	(a)	(b)	(c)	(d)	(e)	(f)
Theoretical *K* value	0.5	0.7	1.0	1.0	2.0	2.0
Recommended design value when ideal conditions are approximated	0.65	0.80	1.2	1.0	2.1	2.0

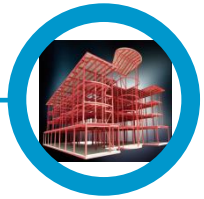


Development of column formulas

- In addition, there are other minor assumptions we haven't covered:
  - Loading is through the centroid
    - Handled through beam/column interaction equations (we'll discuss later!).
  - No twisting/shear deformation
    - Not a major concern for W-shapes.

- However, there are two major reasons why real column behavior deviates from ideal column behavior:

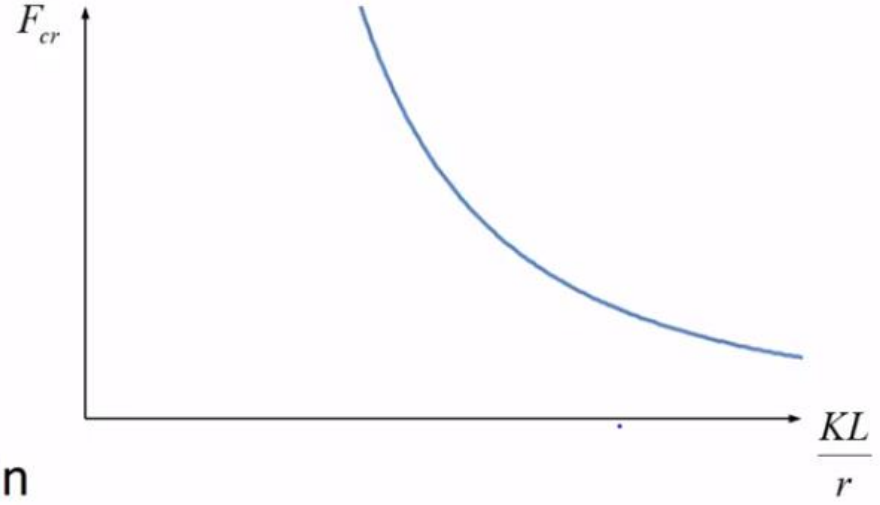
  - Presence of residual stresses
  - Presence of geometric imperfections



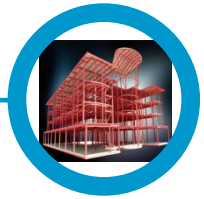
Development of column formulas

- Let's recall the elastic buckling stress of a column.
  - Note the inclusion of "K"

$$F_{cr} = \frac{\pi^2 E}{\left(\frac{KL}{r}\right)^2}$$



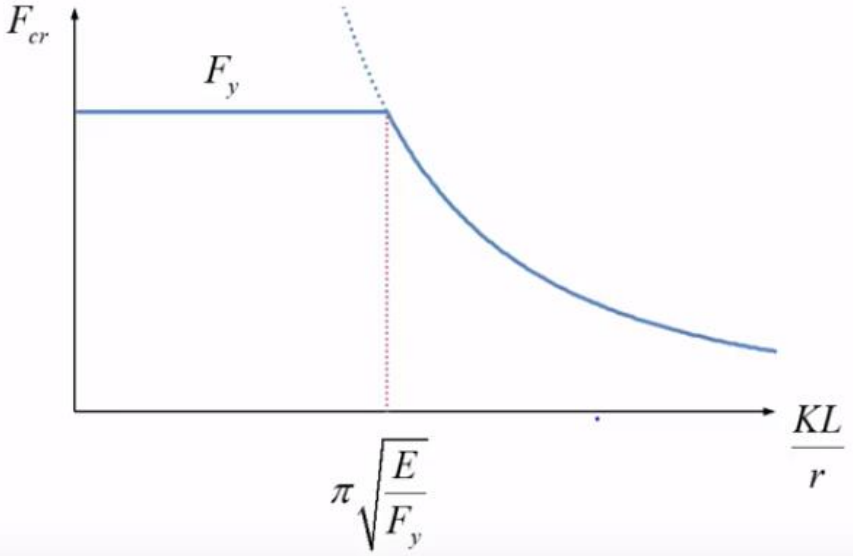
- If we plot this vs. KL/r, we would obtain a hyperbola.

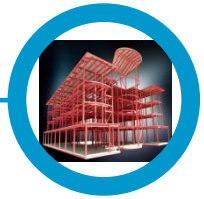


Development of column formulas

- Since this equation is derived based on elastic behavior, the critical stress cannot exceed the yield stress.

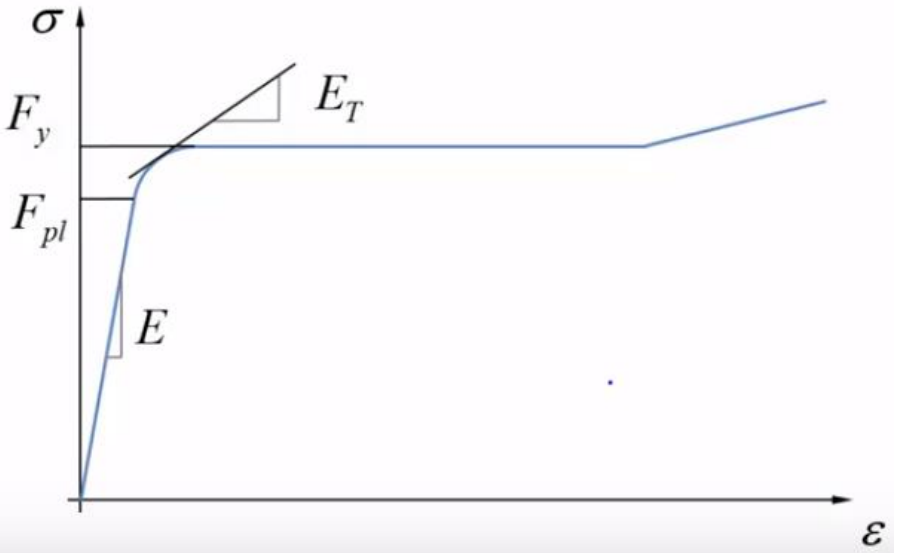
- This cutoff occurs at the ratio shown below:
  - Found by setting the critical stress equal to the yield stress.

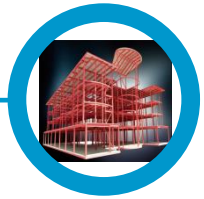




Development of column formulas

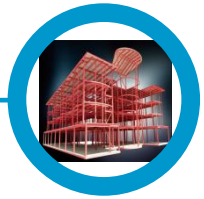
- Between the proportional limit and the yield stress, a region of constrained inelastic behavior is observed.
  - Onset of full yielding.
- In this region, the overall stiffness is reduced.
  - The modulus of elasticity changes as stress increases.
  - Note the presence of a tangent elastic modulus.





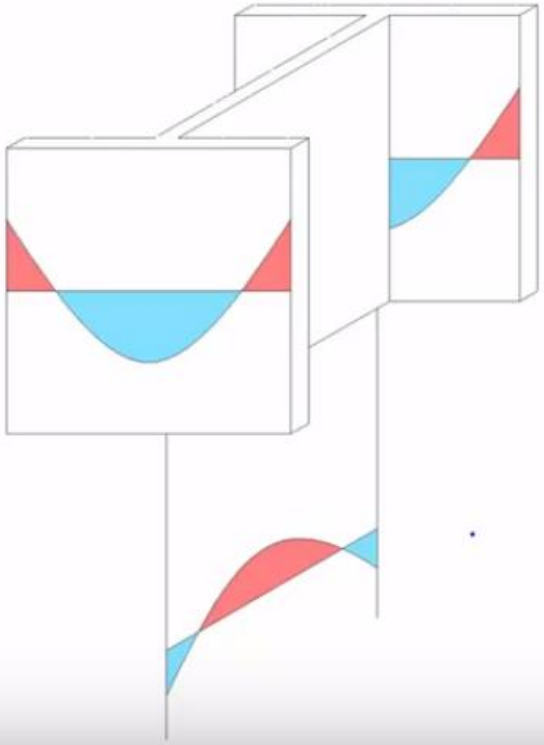
Residual Stresses

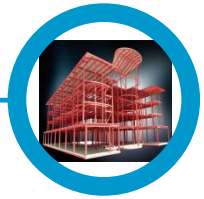
- This transition region wouldn't be a big deal if it weren't for residual stresses.
  - Residual stresses are stresses that remain in a member after it is has been formed into a structural product.
  - Results from:
    - Uneven cooling
    - Cold bending/cambering
    - Welding
- With residual stresses in a column, the onset of yielding occurs much sooner than if no residual stresses are present.
  - Therefore, the effect of residual stresses needs to be captured when analyzing/designing steel compression members.



Development of column formulas

- In rolled shapes, after the hot rolling process:
  - Flange tips and the center of the web (i.e. the exposed regions) cool first, then solidify and shorten.
  - This causes these regions to go into compression.
  - The flange-to-web junction then cools and goes into tension to restore equilibrium in the section.

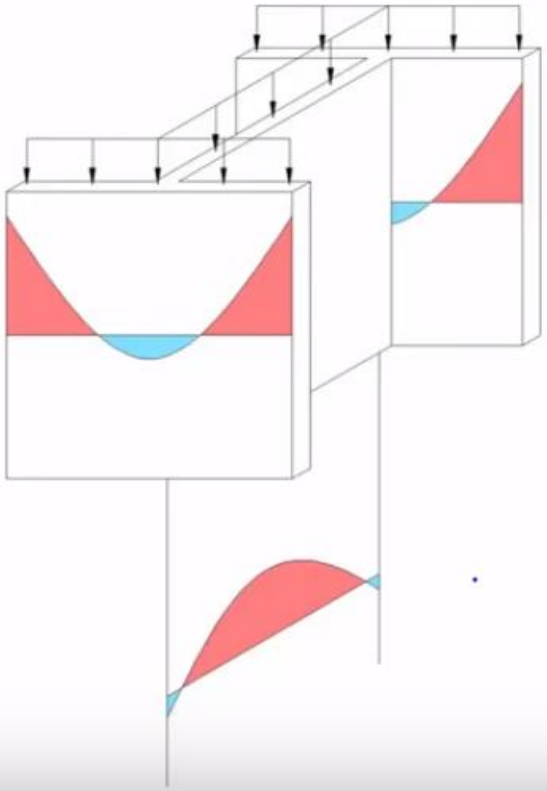


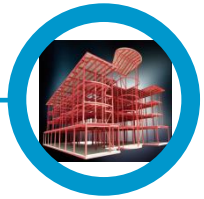


Development of column formulas

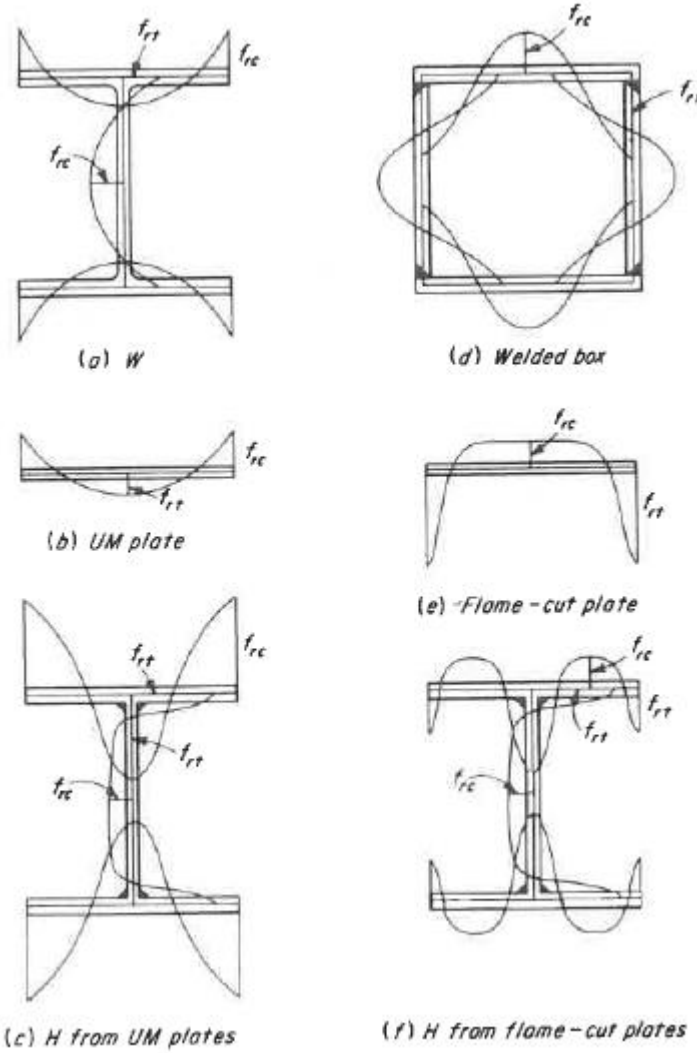
- Wide-flange section loaded with an axial stress:
  - Creates uneven stress distribution.
  - Can cause yielding before the load reaches $F_y A_g$.

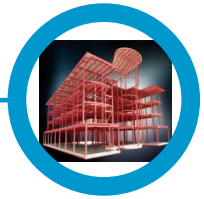
- This effect can have a significant impact on the capacity of steel columns!





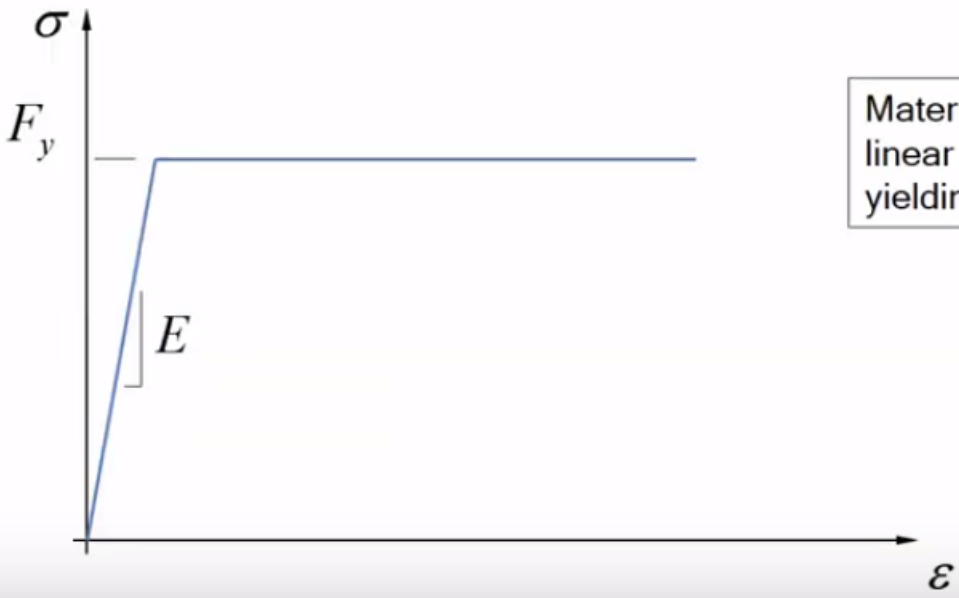
Development of column formulas



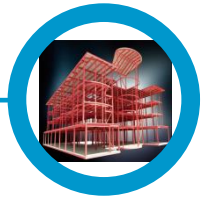


Development of column formulas

- Stress-strain curve (no residual stresses):

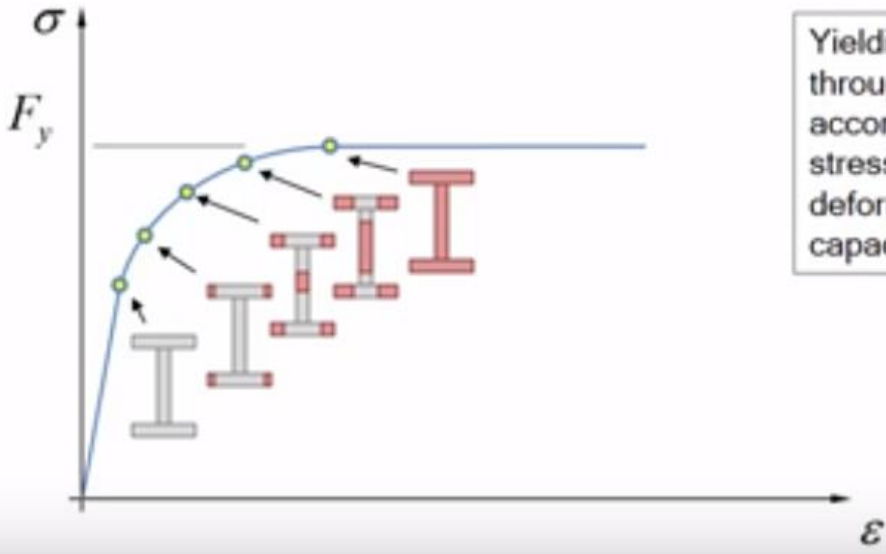


Material will behave in a linear elastic fashion until yielding.

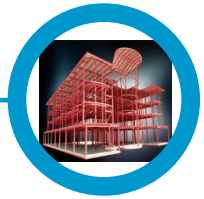


Development of column formulas

- Stress-strain curve (with residual stresses):

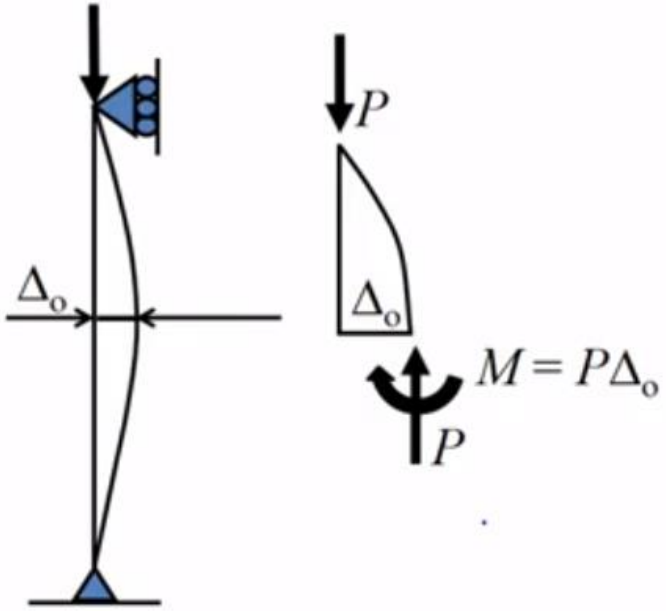


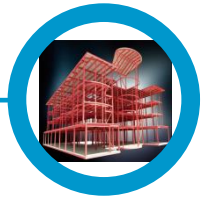
Yielding will progress through the section according to residual stresses, resulting in more deformations and loss in capacity.



Development of column formulas

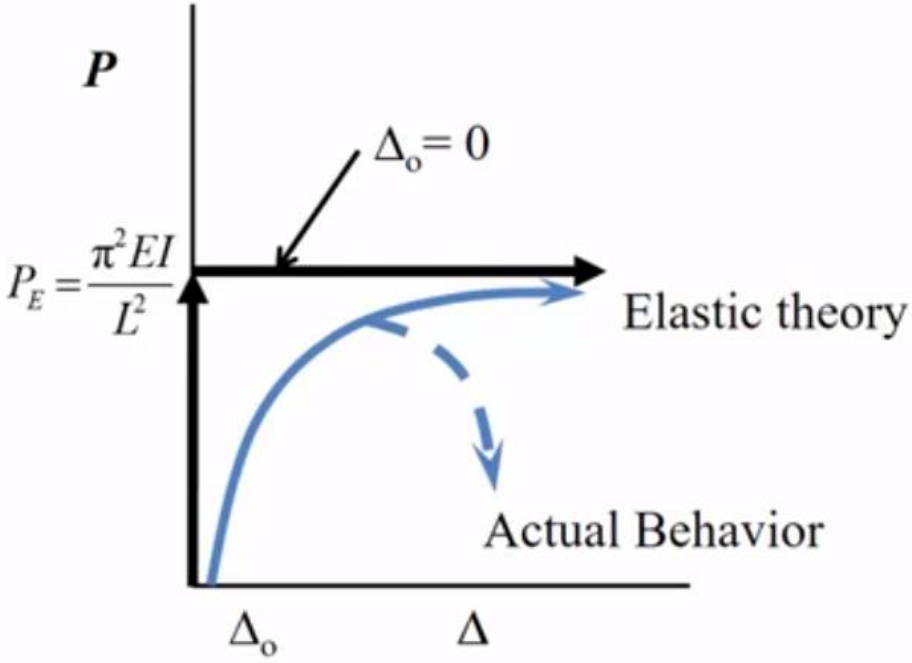
- We derived the Euler buckling capacity based on the assumption that the column was perfectly straight.
  - However, all columns possess some amount of out-of-straightness.
- How does column behavior change if an imperfection is present along the column's length?

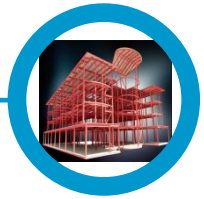




Development of column formulas

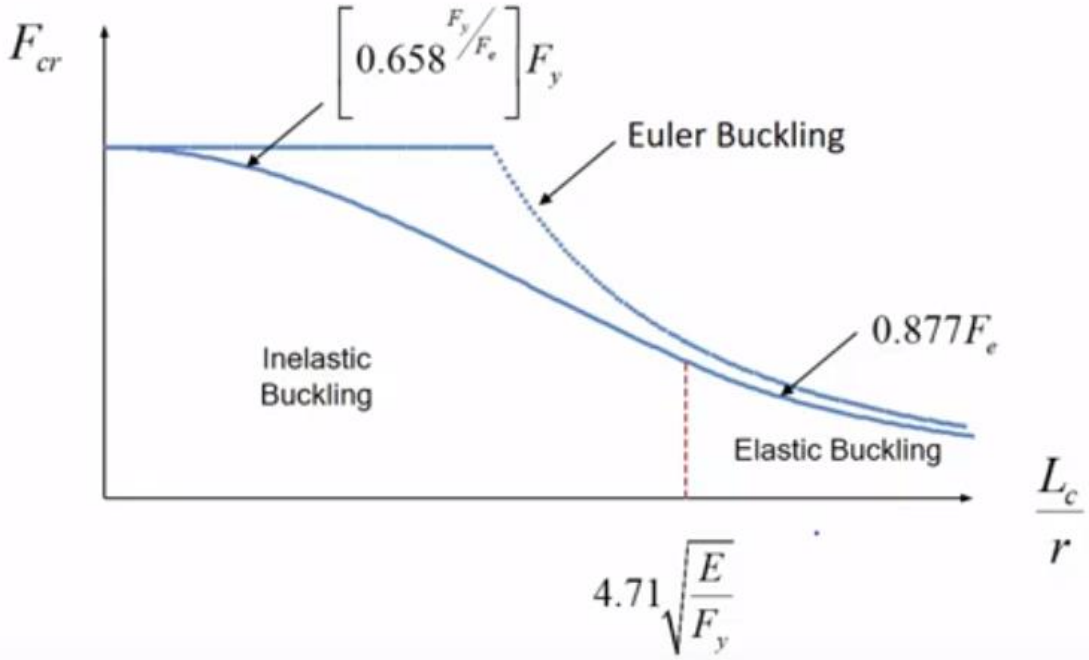
- P-δ curve for an actual column (considering inelasticity / residual stresses):

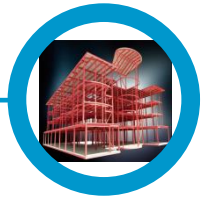




Development of column formulas

- Taking all this into account, AISC adopts the following curve for column capacity:
  - Instead of simply Euler buckling, this curve better takes into account the effects associated with:
    - Residual stresses
    - Geometric imperfections





Development of column formulas

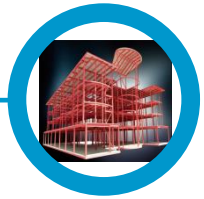
- Capacity of columns is governed by the following equation:

$$P_n = F_{cr} A_g$$
$$(\phi_c = 0.90)$$

Eqn. E3-1: Pg. 16.1-35

  - F_{cr} = critical buckling stress for columns (ksi)
  - A_g = gross area of the column (in^2)

  - Critical buckling stress is found by:
    - Determining the slenderness of the column.
    - Determining whether inelastic or elastic buckling will occur.



Development of column formulas

- The column critical buckling stress are governed by the following equations:

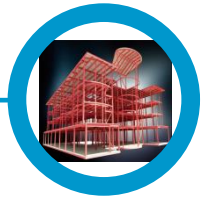
$$\text{if}: \frac{L_c}{r} \le 4.71\sqrt{\frac{E}{F_y}} \qquad F_{cr} = \left[0.658^{\frac{F_y}{F_e}}\right] F_y$$

Eqn. E3-2, Pg. 16.1-33

$$\text{if}: \frac{L_c}{r} > 4.71\sqrt{\frac{E}{F_y}} \qquad F_{cr} = 0.877 F_e$$

- L_c/r = effective column slenderness
- E = steel modulus of elasticity = 29000 ksi
- F_y = yield stress (ksi)
- F_e = elastic/Euler buckling stress (ksi)

Note:
L_c = KL = Effective length of column.



Development of column formulas

- The elastic buckling stress is calculated as follows:

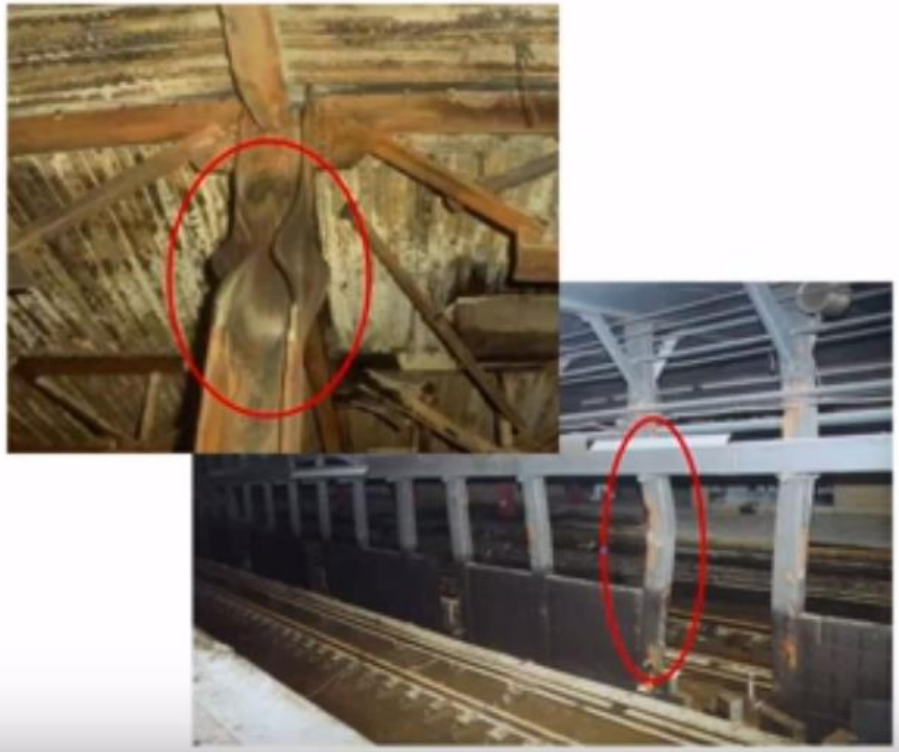
$$F_e = \frac{\pi^2 E}{(L_c/r)^2} = \frac{\pi^2 E}{(KL/r)^2}$$

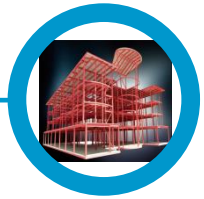
Eqn. E3-4,

  - K = effective length factor
  - L = column length in inches (PAY ATTENTION TO UNITS)
  - r = radius of gyration (in)
  - E = steel modulus of elasticity = 29000 ksi

Local Buckling

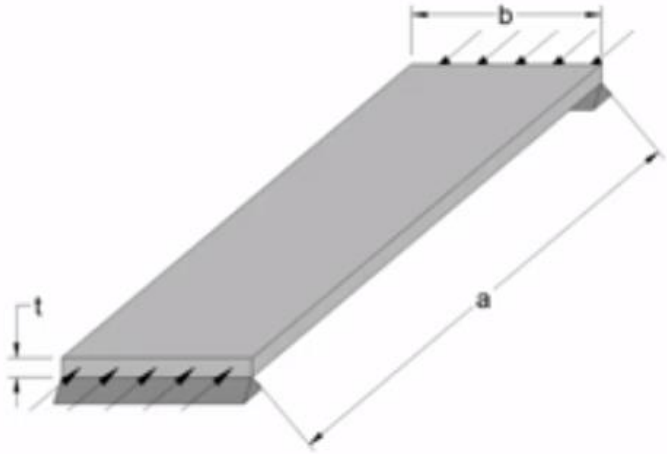
- Column strength based on overall slenderness can only be achieved if the individual plates don't buckle first.
  - If global buckling occurs before local buckling
- Each element of the cross-section can be modeled as a plate supported on one/both edges.
  - Such as flanges and webs of W-sections



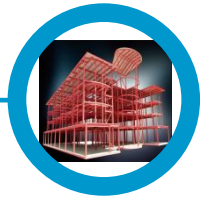


Local Buckling

- Much like column buckling capacity, the buckling capacity of a plate can be derived.
  - The equations are a little more complex, but the process is still the same.
    - The elastic buckling capacity of a plate strip is computed as follows:

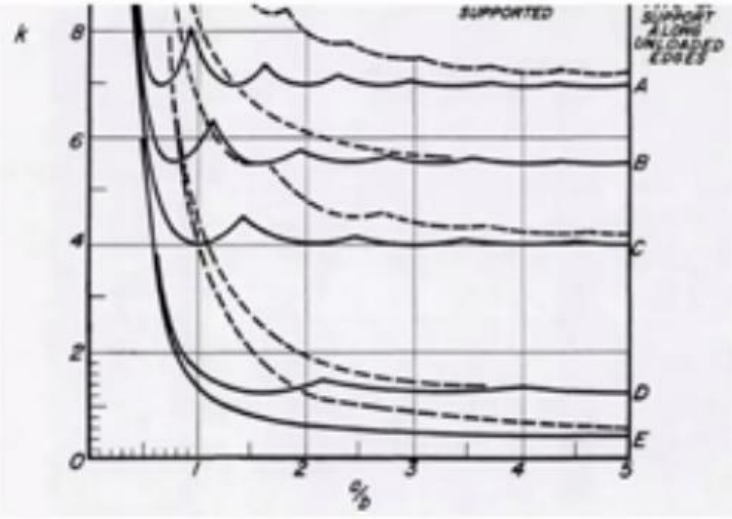


$$F_{cr} = k\frac{\pi^2 E}{12\left(1-\nu^2\right)\left(b/t\right)^2}$$



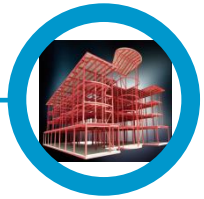
Local Buckling

- The minimum "k" values are as follows:
  - Fixed-fixed: 6.970 (Case A)
  - Fixed-pinned: 5.420 (Case B)
  - Pinned-pinned: 4.000 (Case C)
  - Fixed-free: 1.277 (Case D)
  - Pinned-free: 0.425 (Case E)



- Where do these "k" values come into play?
  - Let's look back at the critical stress equation.

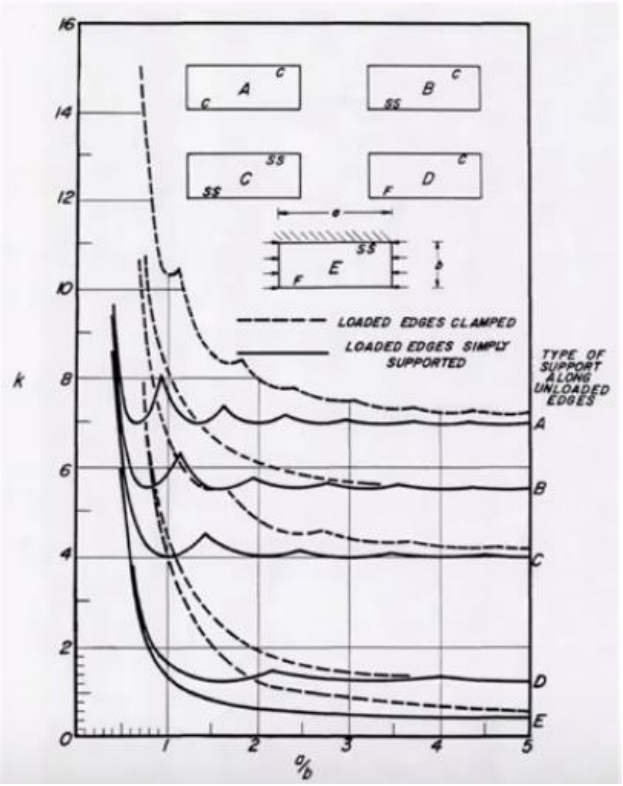
$$F_{cr} = k\frac{\pi^2 E}{12\left(1-\nu^2\right)\left(b/t\right)^2}$$

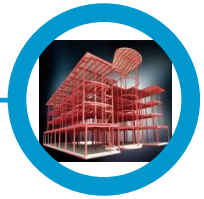


Local Buckling

- The coefficient “k” is much like the effective length factor, “K” for columns.
  - However, as the dimensions and the boundary conditions of the plate change, “k” can change dramatically.

- For each possible iteration, values of “k” are shown:
  - Changing the dimensions of the plate
  - Changing the boundary conditions.

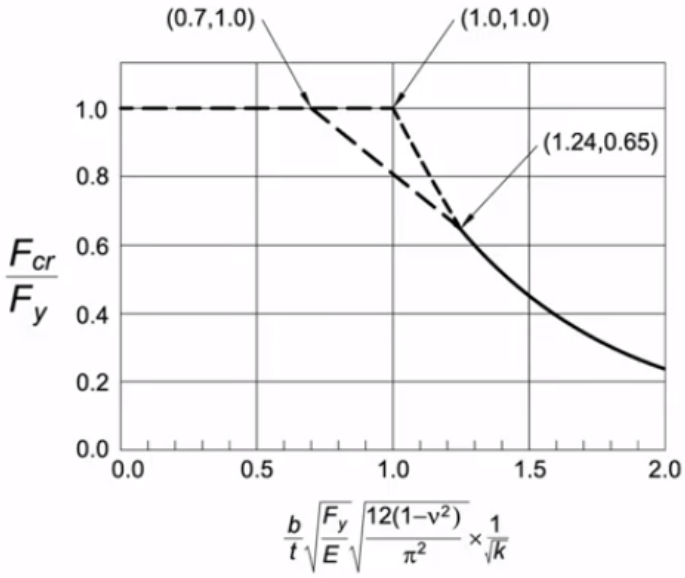


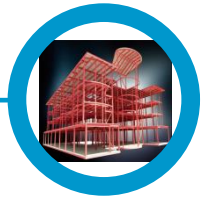


Local Buckling

- Like columns, plates have inelastic buckling capacity, and can be plotted as shown:
  - Transition instead occurs when limit reaches 0.7

$$\frac{b}{t}\sqrt{\frac{F_y}{E}}\sqrt{\frac{12\left(1-\nu^2\right)}{\pi^2}}\frac{1}{\sqrt{k}} \le 0.7$$





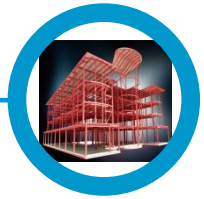
Local Buckling

- The Specification limits local buckling stresses are limited to the yield stress:
  - Basically, dictating that yielding must occur first.

$$F_y \le k \frac{\pi^2 E}{12\left(1-\nu^2\right)(b/t)^2}$$

- With a little bit of algebra, we can rearrange this equation as follows:

$$\frac{b}{t}\sqrt{\frac{F_y}{E}}\sqrt{\frac{12\left(1-\nu^2\right)}{\pi^2}}\frac{1}{\sqrt{k}} \le 1.0$$



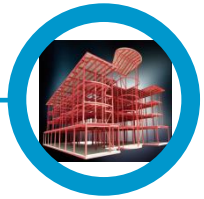
Local Buckling

- With a little more algebra, we can rearrange this equation yet again to solve for the width/thickness ratio, b/t:

$$\frac{b}{t}\sqrt{\frac{F_y}{E}}\sqrt{\frac{12\left(1-\nu^2\right)}{\pi^2}}\frac{1}{\sqrt{k}} \le 1.0 \qquad \therefore \frac{b}{t} \le 0.7\sqrt{\frac{\pi^2}{12\left(1-\nu^2\right)}}\sqrt{k}\sqrt{\frac{E}{F_y}}$$

  - Recognizing that the Poisson's Ratio for steel, v = 0.30, we can simplify our expression a bit:

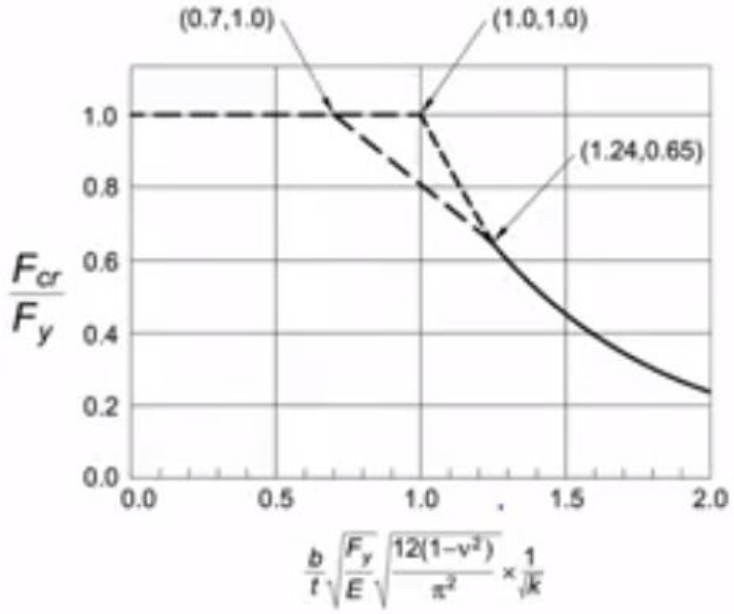
$$\frac{b}{t} \le 0.665\sqrt{k}\sqrt{\frac{E}{F_y}}$$

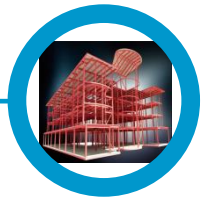


Local Buckling

- Like columns, plates have inelastic buckling capacity, and can be plotted as shown:
  - Transition instead occurs when limit reaches 0.7

$$\frac{b}{t}\sqrt{\frac{F_y}{E}}\sqrt{\frac{12\left(1-\nu^2\right)}{\pi^2}}\frac{1}{\sqrt{k}} \le 0.7$$

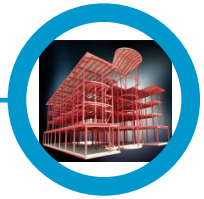




Local Buckling

- What we've just done is derive a limiting width-thickness ratio for plate elements.
  - This will allow us to easily determine whether local buckling will occur or not.
    - Now, we just need to come up with some limiting values.

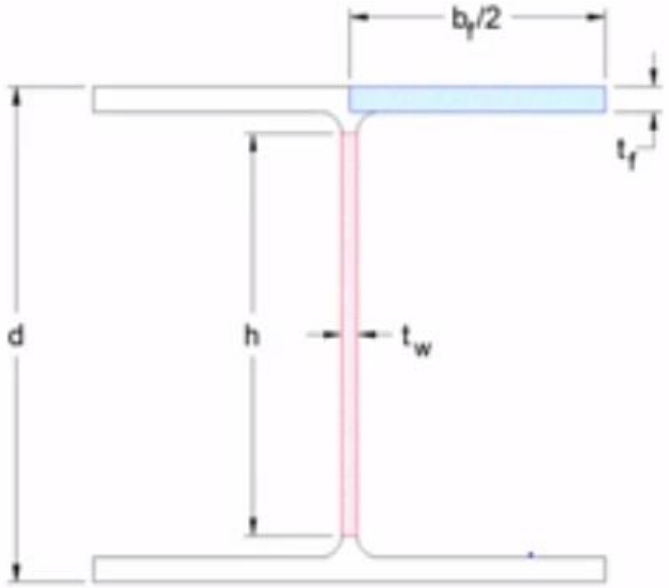
- To derive limits, we need to differentiate between two types of plate elements:
  - Stiffened elements - supported on both edges
    - Webs of I-sections
  - Unstiffened elements - supported on one edge
    - Flanges of I-sections
    - Angle legs
    - Channel flanges
    - Stems of T-sections

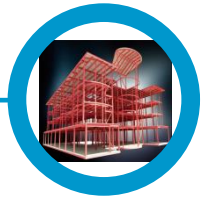


Local Buckling

- For these elements, we need to determine a "k" value that's suitable.
  - For the flanges and webs of rolled I-sections, we assume ⅓ of the way between fixed and pinned.
  - This will help us determine "k."

$$\frac{b}{t} \le 0.665\sqrt{k}\sqrt{\frac{E}{F_y}}$$





Local Buckling

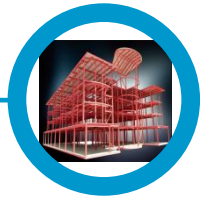
- Therefore:

$$\frac{b}{t} \le 0.665\sqrt{k}\sqrt{\frac{E}{F_y}}$$

  - Unstiffened elements:
    - Fixed-free: 1.277 (Case D)
    - Pinned-free: 0.425 (Case E)
    - Therefore, we'll assume k ≈ 0.70
  - Stiffened elements:
    - Fixed-fixed: 6.970 (Case A)
    - Pinned-pinned: 4.000 (Case C)
    - Therefore, we'll assume k ≈ 5.00

$$\left(b/t\right)_{\text{unstiffened}} \le 0.557\sqrt{\frac{E}{F_y}}$$

$$\left(b/t\right)_{\text{stiffened}} \le 1.488\sqrt{\frac{E}{F_y}}$$

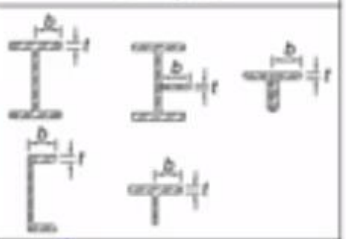
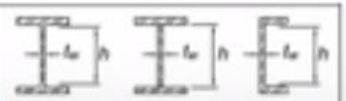


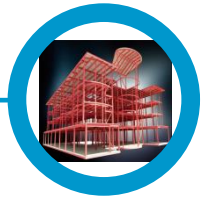
Local Buckling

Pg. 16.1-17

- These limits (and many like it) are defined in the Specifications:
  - *Tab. B4.1a* (columns)
  - *Tab. B4.1b* (beams)
- If a W-section meets these limits, it's said to be a compact section, or a section with compact elements.
  - If a W-section fails these limits, it is said to have slender elements.

TABLE B4.1a
Width-to-Thickness Ratios: Compression Elements Members Subject to Axial Compression

Case	Description of Element	Width-to-Thickness Ratio	Limiting Width-to-Thickness Ratio λ_r (nonslender/slender)	Examples
1	Flanges of rolled I-shaped sections, plates projecting from rolled I-shaped sections, outstanding legs of pairs of angles connected with continuous contact, flanges of channels, and flanges of tees	b/t	$0.56\sqrt{\frac{E}{F_y}}$	
5	Webs of doubly symmetric rolled and built-up I-shaped sections and channels	h/t_w	$1.49\sqrt{\frac{E}{F_y}}$	



Local Buckling

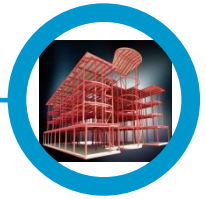
- The first step is to determine whether the section contains compact elements:
  - Slenderness limits: *Tab. B4.1a*

$$\lambda_f \le \lambda_{rf} \qquad \frac{b_f}{2t_f} \le 0.56\sqrt{\frac{E}{F_y}}$$

$$\lambda_w \le \lambda_{rw} \qquad \frac{h}{t_w} \le 1.49\sqrt{\frac{E}{F_y}}$$

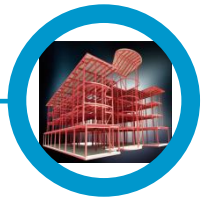
  - Note: If these limits were violated, it wouldn't mean that the columns would be unusable. We'd just have to use a different section of the Specifications:
    - *Sec. E3: Flexural Buckling of Members without Slender Elements*
    - *Sec. E7: Members with Slender Elements*

Compression Members



- Once we've verified that the section is compact is applicable, we have three options for determining the column capacity.
  - Method 1: AISC Column Curve Equations
  - Method 2: Tab. 4-1 (Design Aid)
  - Method 3: Tab. 4-22 (Design Aid)

Compression Members



- The first method simply involves directly computing column capacity:

$$\boxed{\phi_c P_n = 0.90 F_{cr} A_g}$$

$$F_{cr} = \begin{cases} \left[0.658^{\frac{F_y}{F_e}}\right] F_y & \text{if : } \dfrac{KL}{r} \le 4.71\sqrt{\dfrac{E}{F_y}} \\ 0.877 F_e & \text{if : } \dfrac{KL}{r} > 4.71\sqrt{\dfrac{E}{F_y}} \end{cases}$$

$$F_e = \frac{\pi^2 E}{(KL/r)^2}$$

Compression Members



- The first aid we'll employ lists the capacity as a function of the column's effective length.
  - Only works for 50-ksi steel, though.
- This will be a very important tool for design!

Pg. 4-12 ★

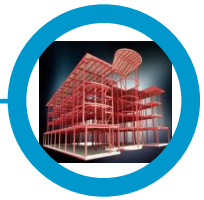
Table 4-1 (continued)
Available Strength in Axial Compression, kips
W-Shapes

W10

F_y = 50 ksi

Shape		W10×									
lb/ft		54		49		45		39		33	
Design		P_n/Ω_c	$\phi_c P_n$	P_n/Ω_c	$\phi_c P_n$	P_n/Ω_c	$\phi_c P_n$	P_n/Ω_c	$\phi_c P_n$	P_n/Ω_c	$\phi_c P_n$
		ASD	LRFD	ASD	LRFD	ASD	LRFD	ASD	LRFD	ASD	LRFD
least radius of gyration, r_y	0	473	711	431	648	398	598	344	517	291	437
	6	446	671	407	611	363	545	313	470	263	395
	7	437	657	398	598	350	527	302	454	253	381
	8	427	642	388	584	337	507	290	436	243	365
	9	415	624	378	568	322	485	277	416	232	348
	10	403	605	366	550	307	461	263	396	220	330
	11	389	585	354	532	291	437	249	374	207	311
	12	375	564	341	512	274	411	234	352	194	292
	13	361	542	327	492	256	385	219	329	181	272
	14	345	519	313	471	239	359	203	306	168	253
	15	330	495	299	449	222	333	188	283	155	233

Compression Members



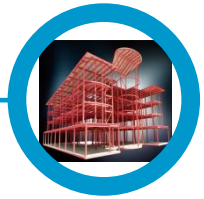
Pg. 4-322 ★

- This aid lists the critical stress as a function of the column's effective length.
  - Much more versatile for other steel grades.
  - This serves as a great "gutcheck" to see if your calculations were correct.
  - Also, ϕ_c is already incorporated!

Table 4-22 (continued)
Available Critical Stress for Compression Members

F_y = 35 ksi			F_y = 36 ksi			F_y = 42 ksi			F_y = 46 ksi			F_y = 50 ksi		
$\frac{KL}{r}$	F_{cr}/Ω_c ksi	$\phi_c F_{cr}$ ksi	$\frac{KL}{r}$	F_{cr}/Ω_c ksi	$\phi_c F_{cr}$ ksi	$\frac{KL}{r}$	F_{cr}/Ω_c ksi	$\phi_c F_{cr}$ ksi	$\frac{KL}{r}$	F_{cr}/Ω_c ksi	$\phi_c F_{cr}$ ksi	$\frac{KL}{r}$	F_{cr}/Ω_c ksi	$\phi_c F_{cr}$ ksi
	ASD	LRFD		ASD	LRFD		ASD	LRFD		ASD	LRFD		ASD	LRFD
81	15.0	22.5	81	15.3	22.9	81	16.8	25.3	81	17.7	26.6	81	18.5	27.9
82	14.9	22.3	82	15.1	22.7	82	16.6	25.0	82	17.5	26.3	82	18.3	27.5
83	14.7	22.1	83	15.0	22.5	83	16.5	24.8	83	17.3	26.0	83	18.1	27.2
84	14.6	22.0	84	14.9	22.3	84	16.3	24.5	84	17.1	25.8	84	17.9	26.9
85	14.5	21.8	85	14.7	22.1	85	16.1	24.3	85	16.9	25.5	85	17.7	26.5
86	14.4	21.6	86	14.6	22.0	86	16.0	24.0	86	16.7	25.2	86	17.4	26.2
87	14.2	21.4	87	14.5	21.8	87	15.8	23.7	87	16.6	24.9	87	17.2	25.9
88	14.1	21.2	88	14.3	21.6	88	15.6	23.5	88	16.4	24.6	88	17.0	25.5
89	14.0	21.0	89	14.2	21.4	89	15.5	23.2	89	16.2	24.3	89	16.8	25.2
90	13.8	20.8	90	14.1	21.2	90	15.3	23.0	90	16.0	24.0	90	16.6	24.9

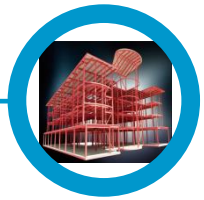
Compression Members



- As shown, the capacity of a column is a function of the radius of gyration.
  - For W-sections, there are two radii of gyration:
    - r_x = radius of gyration about the strong axis
    - r_y = radius of gyration about the weak axis
  - They're referred to this way because r_x is always bigger than r_y.

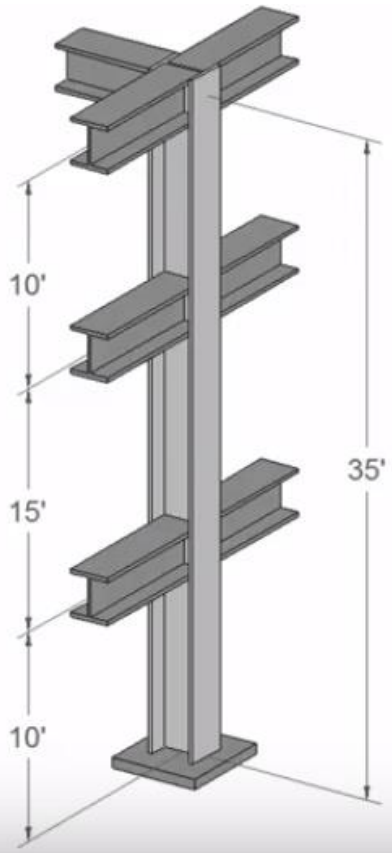
- This raises a question:
  - Do we ever have to check strong-axis buckling capacity in a W-section?

Compression Members

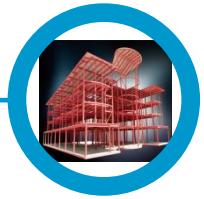


- The answer is yes!
  - The reason why is because while $r_x > r_y$, it doesn't mean that $L_{cx}/r_x > L_{cy}/r_y$

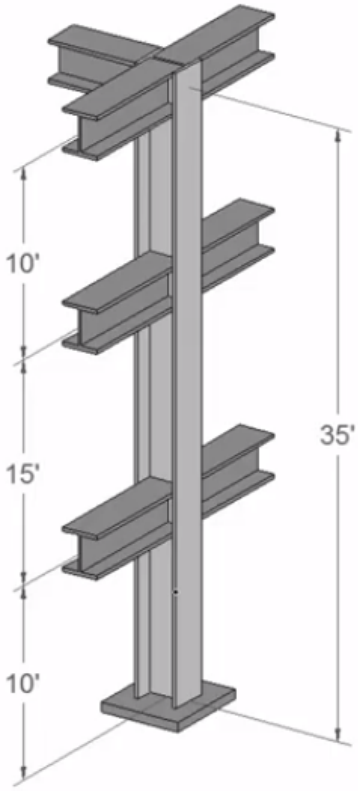
- Let's look at a demonstration:
  - Here the strong axis is unbraced for a length of 35 feet.
  - The weak axis, however, contains intermediate braces.
    - Splitting the weak-axis length into 3 unbraced segments.

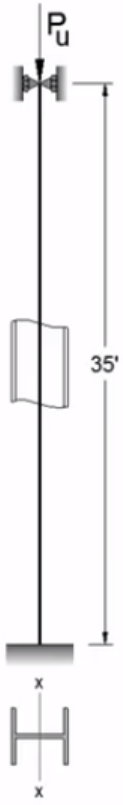


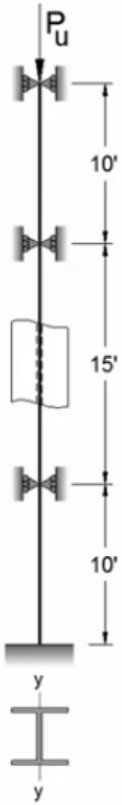
Compression Members

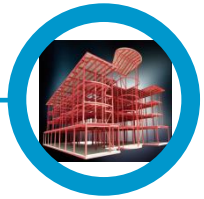


- Analytically, we'd break the column into it's individual segments to determine the governing KL/r.
  - Remember, the larger the KL/r, the weaker the column.



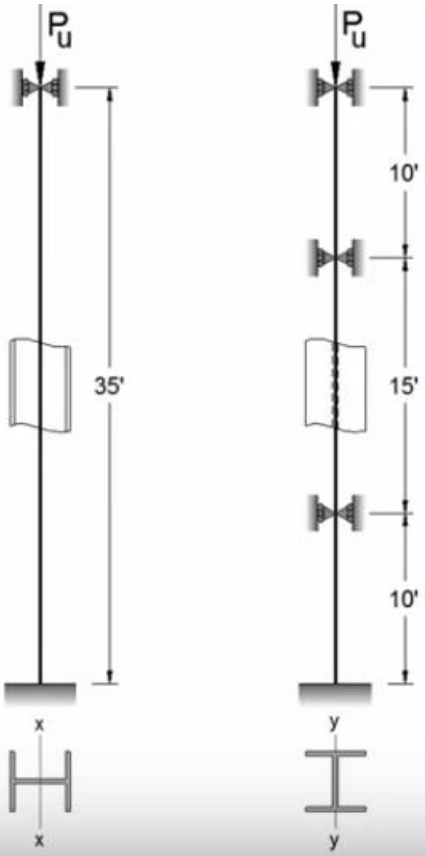






Local Buckling

- Analyze the following column using Method 1.
  - W14×90 (F_y = 50 ksi)
  - The strong-axis bracing is shown on the left.
  - The weak-axis bracing is shown on the right.



Compression Members



$F_y = 50$ ksi

W14 x90: [Pg 1-24 & 1-25]

$A_g = 26.5\ in^2$

$r_x = 6.14"$

$r_y = 3.70"$

$\frac{b_f}{2t_f} = 10.2$ $\quad \frac{h}{t_w} = 25.9$

- consider local buckling, (compact section criteria) [Pg 16.1-17]

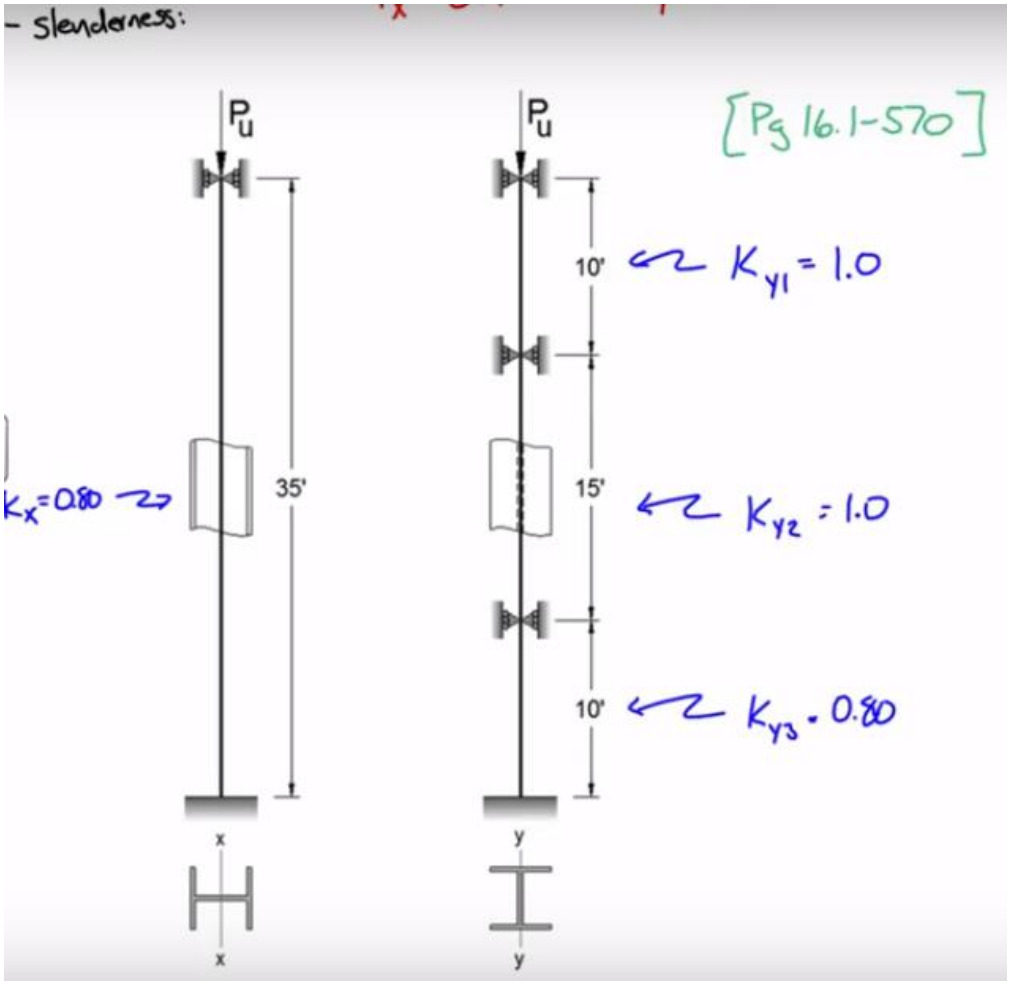
$$\lambda_{rf} = 0.56\sqrt{\frac{E}{F_y}} = 0.56\sqrt{\frac{29000\ ksi}{50\ ksi}} = 13.5$$

$b_f/2t_f = 10.2 \quad \therefore$ Ok

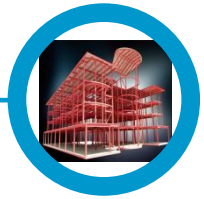
$$\lambda_{rw} = 1.49\sqrt{\frac{E}{F_y}} = 1.49\sqrt{\frac{29000\ ksi}{50\ ksi}} = 35.9$$

$h/t_w = 25.9 \quad \therefore$ Ok

Compression Members



Compression Members



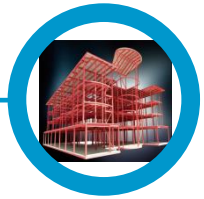
$$\frac{K_x L_x}{r_x} = \frac{(0.80)(35')}{6.14''}\left(12\ ''/ft\right) = 54.723$$

$$\frac{K_{y1} L_{y1}}{r_y} = \frac{(1.0)(10')(12\ ''/ft)}{3.70''} = 32.432$$

$$\frac{K_{y2} L_{y2}}{r_y} = \frac{(1.0)(15')(12\ ''/ft)}{3.70''} = 40.540$$

$$\frac{K_{y3} L_{y3}}{r_y} = \frac{(0.80)(10')(12\ ''/ft)}{3.70''} = 25.950$$

Compression Members



- method 1: Use AISC column curves

$$F_e = \frac{\pi^2 E}{(KL/r)^2} = \frac{\pi^2 (29000\ ksi)}{(54.723)^2} = 95.58\ ksi$$

$$\frac{KL}{r} = 54.723 \qquad 4.71\sqrt{\frac{E}{F_y}} = 4.71\sqrt{\frac{29000\ ksi}{50\ ksi}} = 113.43$$

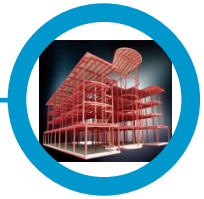
$$\therefore F_{cr} = \left[0.658^{F_y/F_e}\right]F_y = \left[0.658^{50\ ksi/95.58\ ksi}\right](50\ ksi) = 40.17\ ksi$$

$$\phi_c P_n = \phi_c F_{cr} A_g = (0.90)(40.17\ ksi)(26.5\ in^2)$$

ANSWER

$\phi_c P_n = 958.1\ kip$

Compression Members



- The first aid we'll employ lists the capacity as a function of the column's effective length.
  - Only works for 50-ksi steel, though.
- This will be a very important tool for design!

Pg. 4-12 ★



Table 4-1a (continued)

Available Strength in Axial Compression, kips

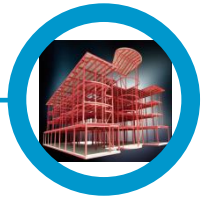
W-Shapes

F_y = 50 ksi

W12

Shape		W12×									
lb/ft		96		87		79		72		65	
Design		P_n/Ω_c	$\phi_c P_n$	P_n/Ω_c	$\phi_c P_n$	P_n/Ω_c	$\phi_c P_n$	P_n/Ω_c	$\phi_c P_n$	P_n/Ω_c	$\phi_c P_n$
		ASD	LRFD	ASD	LRFD	ASD	LRFD	ASD	LRFD	ASD	LRFD
least radius of gyration, r_y	0	844	1270	766	1150	695	1040	632	949	572	859
	6	811	1220	736	1110	667	1000	606	911	549	825
	7	800	1200	726	1090	657	988	597	898	540	812
	8	787	1180	714	1070	646	971	587	883	531	798
	9	772	1160	700	1050	634	953	576	866	521	783
	10	756	1140	685	1030	620	932	564	847	510	766
	11	739	1110	670	1010	606	910	550	827	497	747
	12	720	1080	653	981	590	887	536	806	484	728
	13	701	1050	635	954	574	862	521	783	470	707
	14	680	1020	616	925	556	836	505	759	456	685
	15	659	990	596	896	538	809	489	735	441	663

Compression Members



- This aid lists column capacity based on weak-axis buckling capacity.
  - In other words, it always assumed that weak-axis buckling governs.

$$L_c/r = \frac{L_{cy}}{r_y}$$

  - To check strong-axis buckling capacity, we need to do a conversion:
    - Basically, we set $L_{cy}/r_x = L_{cy}/r_y$ and solve:

$$\frac{L_{cx}}{r_x} = \frac{L_{cy}}{r_y} \qquad \therefore \boxed{L_{cy} = \frac{L_{cx}}{(r_x/r_y)}}$$

Compression Members



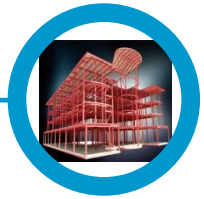
- This aid lists column capacity based on weak-axis buckling capacity.
  - In other words, it always assumed that weak-axis buckling governs.

$$KL/r = \frac{K_y L_y}{r_y}$$

  - To check strong-axis buckling capacity, we need to do a conversion:
    - Basically, we set $(KL/r)_x = (KL/r)_y$ and solve:

$$\frac{K_x L_x}{r_x} = \frac{K_y L_y}{r_y} \qquad \therefore \boxed{K_y L_y = \frac{K_x L_x}{\left(r_x / r_y\right)}}$$

Compression Members

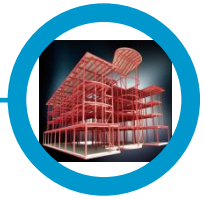


- Therefore, we can check each axis by going into the table with the appropriate value:
  - Weak axis: $KL = K_y L_y$
  - Strong axis: $(KL)_{eff} = \dfrac{K_x L_x}{(r_x / r_y)}$

Properties										
P_{ex}, kips	69.1	104	60.1	90.1	65.3	98.0	54.1	81.1	45.2	67.8
P_{ei}, kips/in.	12.3	18.5	11.3	17.0	11.7	17.5	10.5	15.8	9.67	14.5
P_{eb}, kips	112	168	86.6	130	94.2	142	68.7	103	53.7	80.7
P_{fb}, kips	70.8	106	58.7	88.2	71.9	108	52.6	79.0	35.4	53.2
L_p, ft	9.04		8.97		7.10		6.99		6.85	
L_r, ft	33.6		31.6		26.9		24.2		21.8	
A_g, in.^{2}	15.8		14.4		13.3		11.5		9.71	
I_x, in.^{4}	303		272		248		209		171	
I_y, in.^{4}	103		93.4		53.4		45.0		36.6	
r_y, in.	2.56		2.54		2.01		1.98		1.94	
r_x/r_y	1.71		1.71		2.15		2.16		2.16	
$P_{ex}(KL)^2/10^4$, k-in.^{2}	8670		7790		7100		5980		4890	
$P_{ey}(KL)^2/10^4$, k-in.^{2}	2950		2670		1530		1290		1050	
ASD	**LRFD**		Note: Heavy line indicates KL/r_y equal to or greater than 200.							
$\Omega_c = 1.67$	$\phi_c = 0.90$									

- Also, if you refer to the bottom of the table, r_x/r_y ratios are listed.

Compression Members



- method 2 (Table 4-1a)

- recall: W14x90, $F_y = 50$ ksi

$K_x = 0.80$
$L_x = 35$ ft
} $K_x L_x = 28$ ft

$r_x / r_y = 1.66$

$$(KL)_{equiv} = \frac{K_x L_x}{r_x / r_y} = \frac{28\text{ ft}}{1.66} = 16.87\text{ ft}$$

$\therefore \phi_c P_n (KL = 16') = 979^k$
$\phi_c P_n (KL = 17') = 955^k$

$$\therefore \phi_c P_n = 979^k + \left(\frac{955^k - 979^k}{17' - 16'}\right)(16.87' - 16') = 958.12\text{ kip}$$

ANSWER

$\phi_c P_n = 958.12$ kip

Compression Members

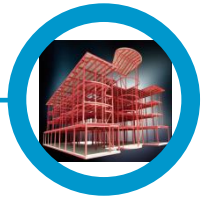


Table 4-22, pg 4-322

- This aid lists the critical stress as a function of the column's effective length.
  - Much more versatile for other steel grades.
  - This serves as a great "gutcheck" to see if your calculations were correct.
  - Also, ϕ_c is already incorporated!

Available Critical Stress for Compression Members

$\frac{L_c}{r}$	$F_y = 35$ ksi		$F_y = 36$ ksi		$F_y = 46$ ksi		$F_y = 50$ ksi		$F_y = 65$ ksi		$F_y = 70$ ksi	
	F_{cr}/Ω_c ksi ASD	$\phi_c F_{cr}$ ksi LRFD	F_{cr}/Ω_c ksi ASD	$\phi_c F_{cr}$ ksi LRFD	F_{cr}/Ω_c ksi ASD	$\phi_c F_{cr}$ ksi LRFD	F_{cr}/Ω_c ksi ASD	$\phi_c F_{cr}$ ksi LRFD	F_{cr}/Ω_c ksi ASD	$\phi_c F_{cr}$ ksi LRFD	F_{cr}/Ω_c ksi ASD	$\phi_c F_{cr}$ ksi LRFD
41	19.2	28.9	19.7	29.7	24.6	37.0	26.5	39.8	33.2	49.9	35.3	53.0
42	19.2	28.8	19.6	29.5	24.5	36.8	26.3	39.5	32.9	49.5	35.0	52.6
43	19.1	28.7	19.6	29.4	24.3	36.6	26.2	39.3	32.6	49.1	34.7	52.1
44	19.0	28.5	19.5	29.3	24.2	36.3	26.0	39.1	32.4	48.7	34.4	51.7
45	18.9	28.4	19.4	29.1	24.0	36.1	25.8	38.8	32.1	48.3	34.1	51.2
46	18.8	28.3	19.3	29.0	23.9	35.9	25.6	38.5	31.8	47.8	33.8	50.7
47	18.7	28.1	19.2	28.9	23.8	35.7	25.5	38.3	31.6	47.4	33.4	50.3
48	18.6	28.0	19.1	28.7	23.6	35.4	25.3	38.0	31.3	47.0	33.1	49.8
49	18.5	27.9	19.0	28.5	23.4	35.2	25.1	37.7	31.0	46.6	32.8	49.3
50	18.4	27.7	18.9	28.4	23.3	35.0	24.9	37.5	30.7	46.1	32.5	48.8

Compression Members

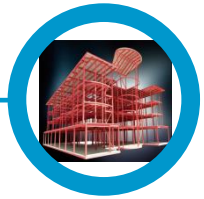


Table 4-22, pg 4-322

- method 3

$KL/r = 54.723$

$\phi_c F_{cr} (KL/r = 54) = 36.4 \text{ ksi}$

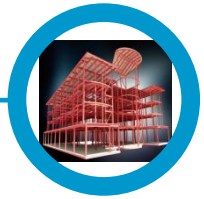
$\phi_c F_{cr} (KL/r = 55) = 36.1 \text{ ksi}$

$\therefore \phi_c F_{cr} (KL/r = 54.723) = 36.183 \text{ ksi}$

$\therefore \phi_c P_n = (36.183 \text{ ksi})(26.5 \text{ in}^2) = 958.9 \text{ kip}$

ANSWER

$\phi_c P_n = 958.9 \text{ k}$



Column Design

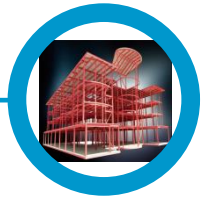
- Procedure:
  - Determine the design load, P_u.
  - Use capacity tables to select trial sections.
  - Selected based on weak-axis buckling capacity.

$$L_c = L_{cy}$$

  - Verify strong-axis buckling capacity.

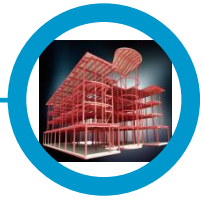
$$(L_c)_{\text{eff}} = \frac{L_{cx}}{(r_x/r_y)}$$

  - Of all valid sections, choose the lightest.



Column Design

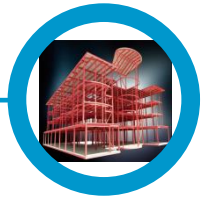
- When selecting sections, be sure and cycle through each family to find the lightest section.
  - In other words, select at least one W14, W12, W10, and W8.
- Also, once a column is selected, its capacity and compactness should be verified!



Column Design

- Select the most economical W-shape column to resist a service dead load of 150 kip and a service live load of 350 kip.
  - Take F_y = 50 ksi

- The column is 15 feet long and pinned on each end in both directions.



Column Design

- factored load: $P_D = 150^k$, $P_L = 350^k$ } $P_u = 1.2P_D + 1.6P_L = 740$ kip

- yield stress: $F_y = 50$ ksi ← PAY ATTENTION!

- effective lengths:
  - 15 ft long
  - pinned in both directions

$K_x L_x = 15$ ft

$K_y L_y = 15$ ft ← only consider this



X-X



Y-Y

- select trial sections:

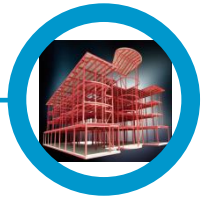
$P_u = 740^k$

$KL_y = 15$ ft

① W14×90 ($\phi_c P_n = 1000^k$)
② W12×79 ($\phi_c P_n = 809^k$)
③ W10×88 ($\phi_c P_n = 831^k$)
④ No W8s

ANSWER

Use a W12×79



Column Design

- factored load: $P_D = 150^k$, $P_L = 350^k$ } $P_u = 1.2P_D + 1.6P_L = 740$ kip

- yield stress: $F_y = 50$ ksi ← PAY ATTENTION!

- effective lengths:
  - 15 ft long
  - pinned in both directions

$K_x L_x = 15$ ft

$K_y L_y = 15$ ft ← only consider this



X-X



Y-Y

- select trial sections:

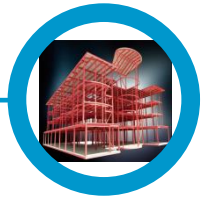
$P_u = 740^k$

$KL_y = 15$ ft

① W14×90 ($\phi_c P_n = 1000^k$)
② W12×79 ($\phi_c P_n = 809^k$)
③ W10×88 ($\phi_c P_n = 831^k$)
④ No W8s

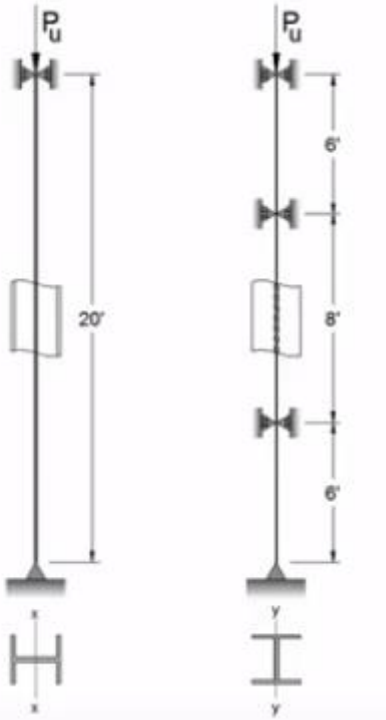
ANSWER

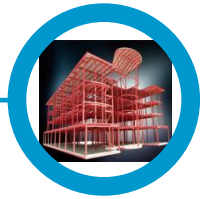
Use a W12×79



Column Design

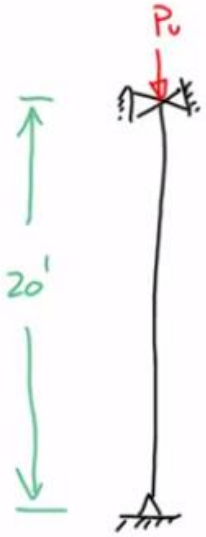
- Select the most economical W-shape column to resist a factored load of 890 kip.
  - Take F_y = 50 ksi
- The column is 20 feet long and bracing conditions about each axis are as shown.

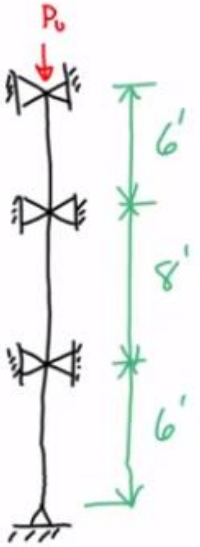




Column Design

- factored load : $P_u = 890\ k$
- $F_y = 50\ ksi$
- effective length :





$K_x L_x = 20\ ft$

$K_y L_y = 8\ ft$

- trial sections :

$P_u = 890k$

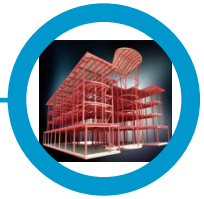
$kL_y = 8\ ft$

① W14×82 ($\phi_c P_n = 968^k$, $r_x/r_y = 2.44$)

② W12×79 ($\phi_c P_n = 971^k$, $r_x/r_y = 1.75$)

③ W10×77 ($\phi_c P_n = 925^k$, $r_x/r_y = 1.73$)

④ No W8s



Column Design

① W14x82: $(KL)_{equiv} = \dfrac{K_x L_x}{r_x/r_y} = \dfrac{20\,ft}{2.44} = 8.20\,ft$

$\phi_c P_n\,(KL = 8\,ft) = 968\,kip$

$\phi_c P_n\,(KL = 9\,ft) = 940\,kip$

$\therefore$ W14x82 is Good

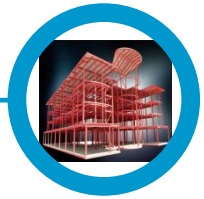
② W12x79 $(KL)_{equiv} = \dfrac{K_x L_x}{r_x/r_y} = \dfrac{20\,ft}{1.75} = 11.43\,ft$

$\phi_c P_n\,(KL = 11\,ft) = 910\,kip$

$\phi_c P_n\,(KL = 12\,ft) = 887\,kip$

$\therefore \phi_c P_n = 910^k + \left(\dfrac{887^k - 910^k}{12' - 11'}\right)(11.43' - 11') = 900.1\,kip$

W12x79 is Good



Column Design

(3) W10x77 $\quad (KL)_{equiv} = \dfrac{K_x L_x}{r_x / r_y} = \dfrac{20\,ft}{1.73} = 11.56\,ft$

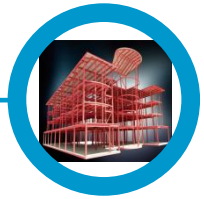
$\phi_c P_n (KL = 11\,ft) = 846\,kip$

$\phi_c P_n (KL = 12\,ft) = 816\,kip$

W10x77 is No Good

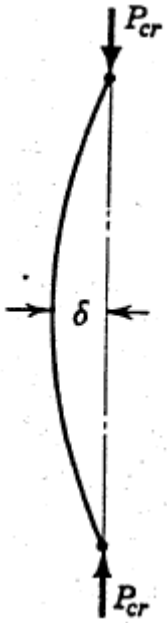
- final design

Use a W12x79 ANSWER



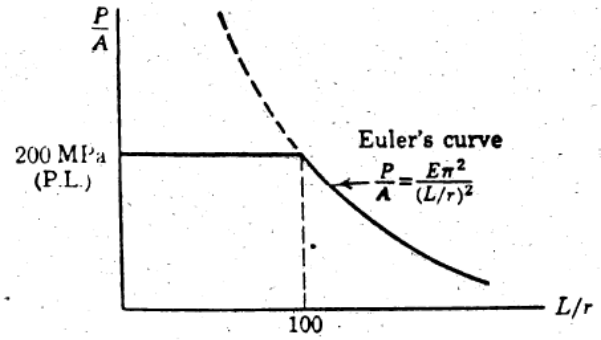
Development of column formulas

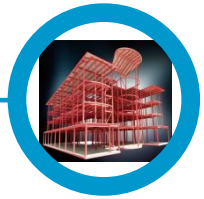
- Critical load is the load required to maintain in its deflected position without any side thrust
- Any increase in P beyond this value increases the deflection
- $M = P_{cr} \times \delta$
- Long columns by Euler's formula
- $P_{cr} = \frac{E\,I\,\pi^2}{L^2}$



$$P = N\frac{EI\pi^2}{L^2} = \frac{EI\pi^2}{L_e^2}$$

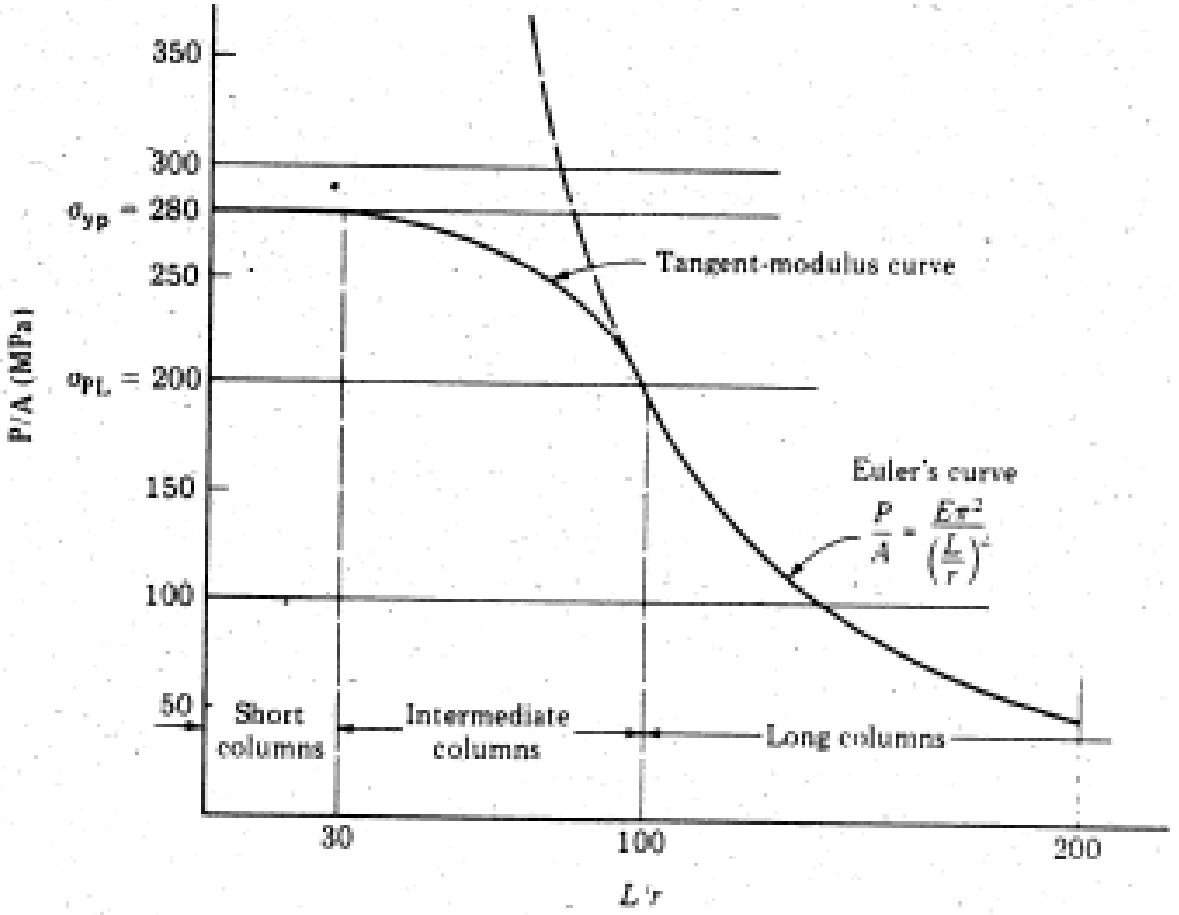
End condition	N = Number of times strength of hinged columns	L_e = Effective length
Fixed ends	4	$\frac{1}{2}L$
One end fixed, the other hinged	2	$0.7L$
Both ends hinged	1	L
One end fixed, the other free	$\frac{1}{4}$	$2L$

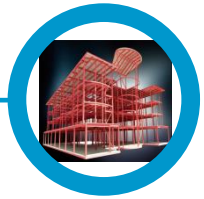




Development of column formulas

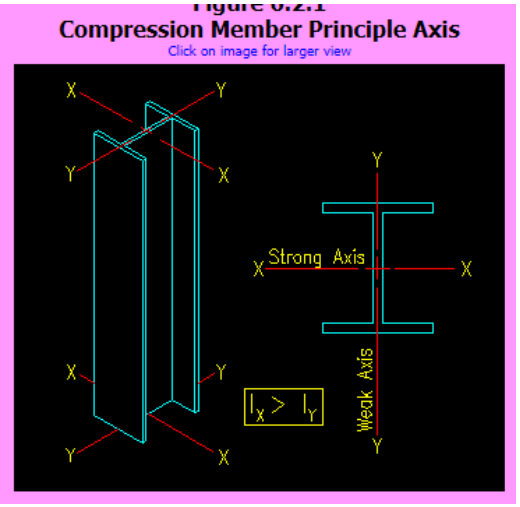
- Euler equation not valid for intermediate and short columns
- Mainly empirical formulas are suggested for intermediate and short columns.





End restraint and effective lengths

- Columns with appreciable rotational and translational end restraint can support more load
- Effective length factor K multiplied with length.
- Important to know whether frames are braced or un-braced



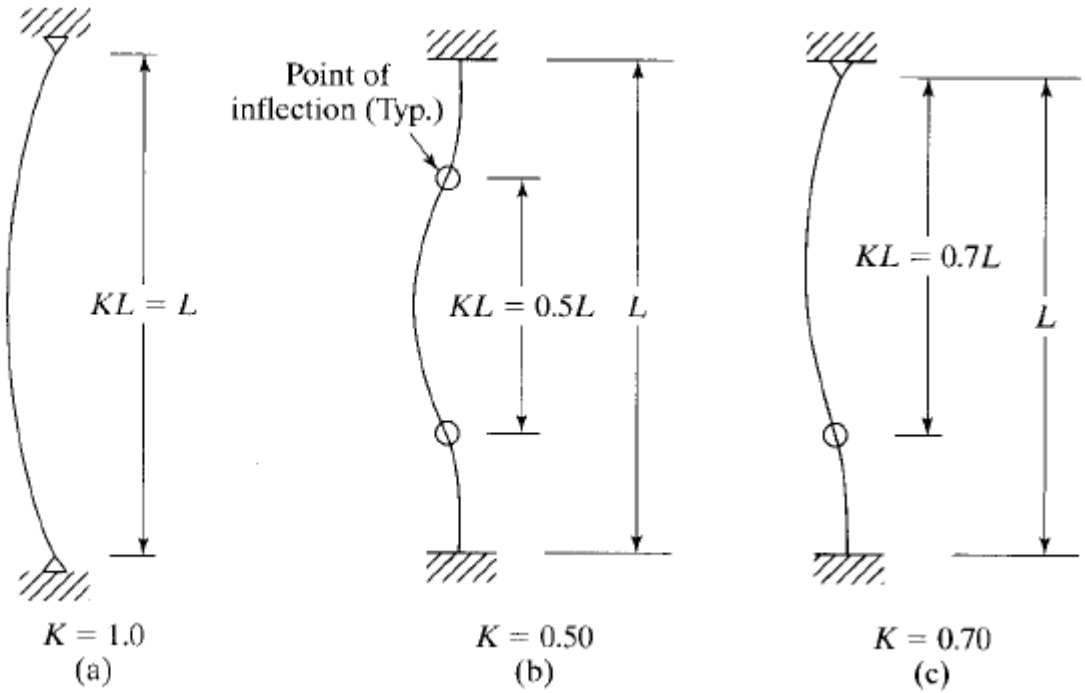
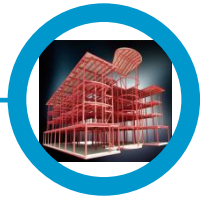


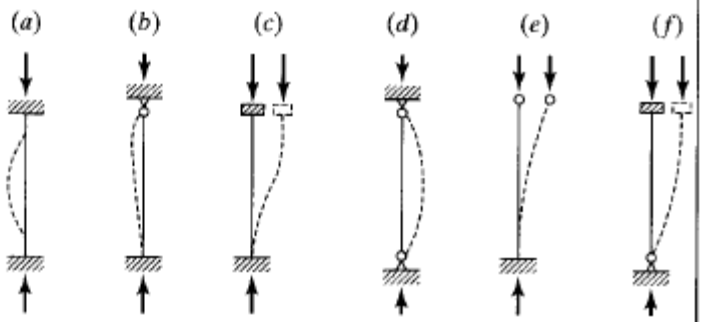
FIGURE 5.4

Effective length (KL) for columns in braced frames (sidesway prevented).



End restraint and effective lengths

TABLE 5.1 Approximate Values of Effective Length Factor, K

Buckled shape of column is shown by dashed line						
	(a)	(b)	(c)	(d)	(e)	(f)
Theoretical K value	0.5	0.7	1.0	1.0	2.0	2.0
Recommended design value when ideal conditions are approximated	0.65	0.80	1.2	1.0	2.10	2.0

End condition code



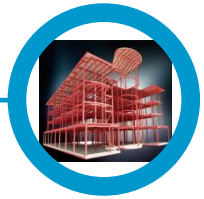
Rotation fixed and translation fixed

Rotation free and translation fixed

Rotation fixed and translation free

Rotation free and translation free

- For continuous frames, use alignment charts.



End restraint and effective lengths

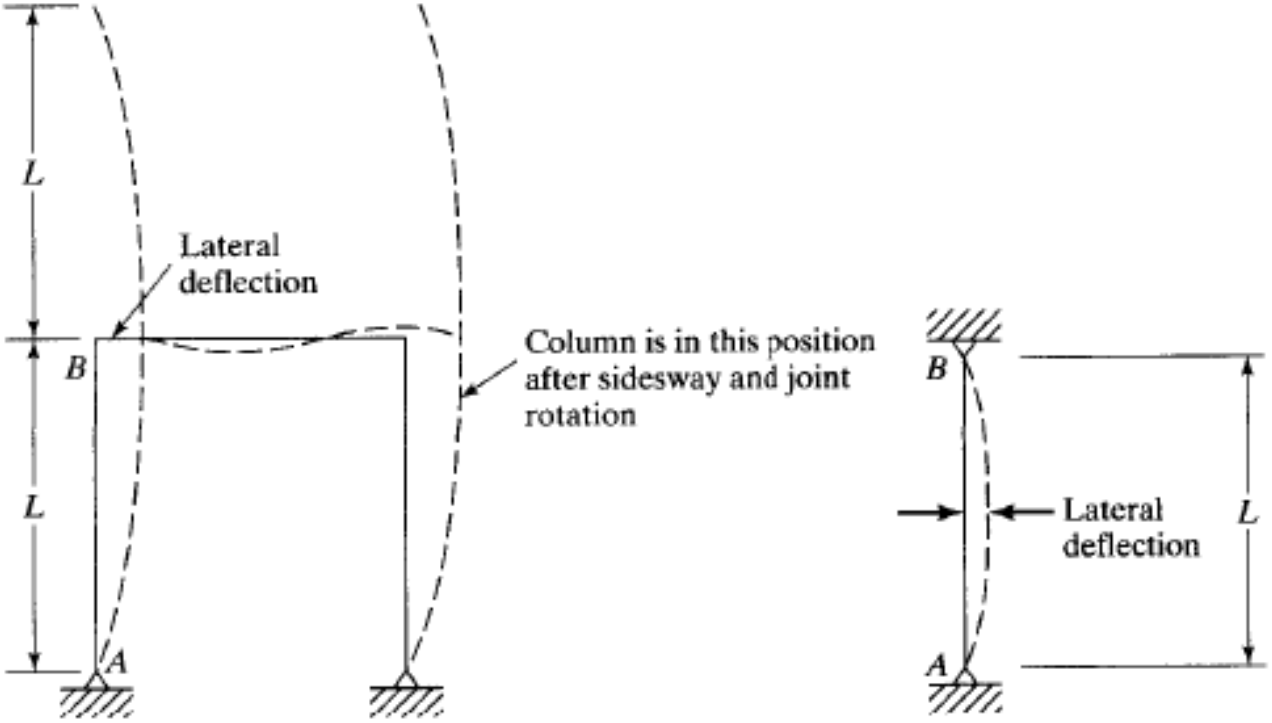


FIGURE 5.5

End restraint and effective lengths

Figure 6.2.1
Compression Member Principle Axis
Click on image for larger view

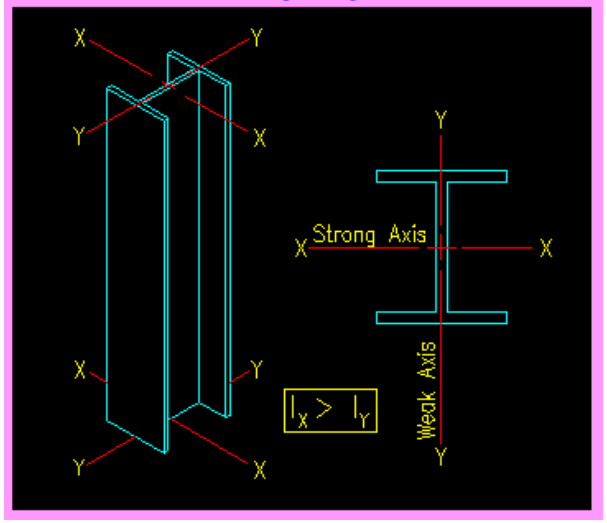
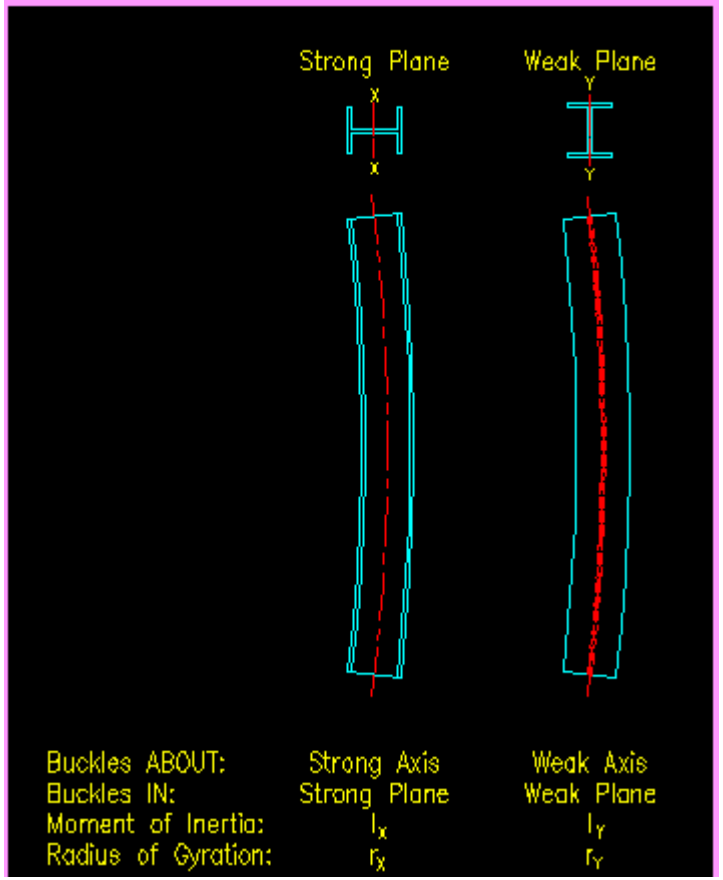
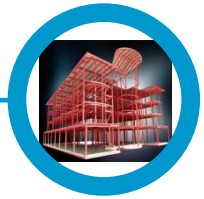


Figure 6.2.2
Compression Member Principle Planes
Click on image for larger view





End restraint and effective lengths

Figure 6.2.3
Braced vs. Unbraced Frames
Click on image for larger view

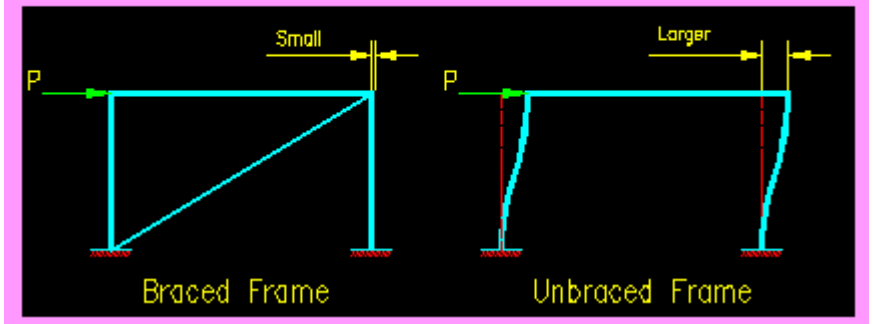
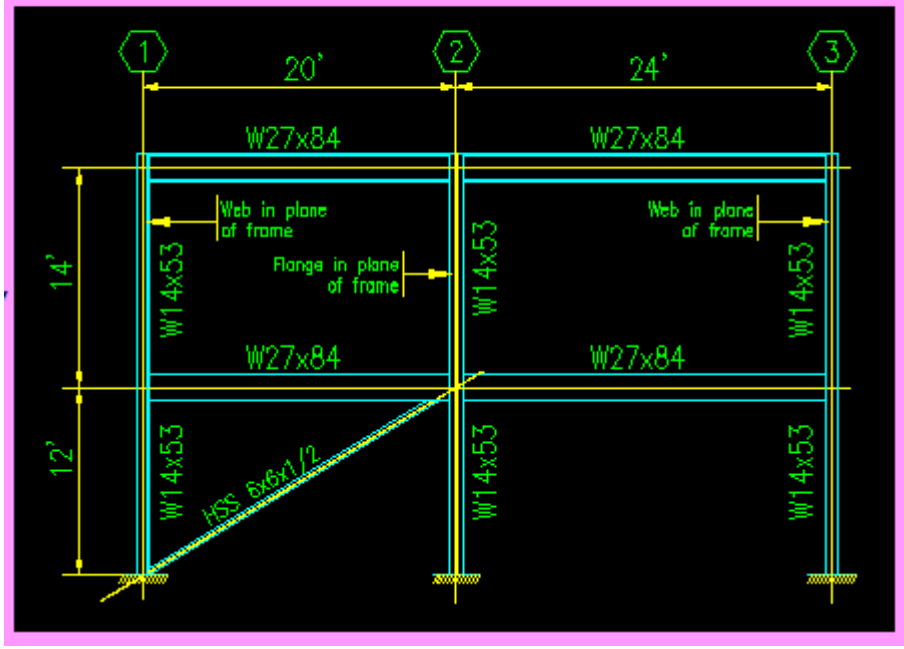
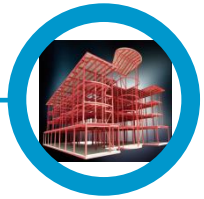


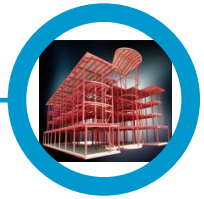
Figure 6.2.4
Bracing by Level
Click on image for larger view





Stiffened and unstiffened elements

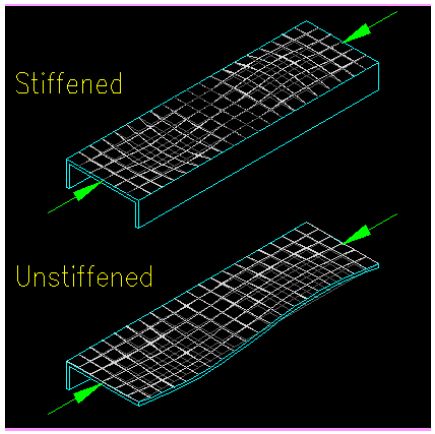
- AISC specification (Section B4) provides limiting values for the width-thickness ratios of the individual parts of compression members
- Unstiffened element is a projecting piece with one free edge parallel to the direction of compression force.
- Stiffened element is supported along the two edges in that direction
- Various range of width to thickness ratio
- AISC specification divides members into three classifications: compact sections, non-compact section and slender compression elements.



Stiffened and unstiffened elements

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Stiffened and unstiffened elements



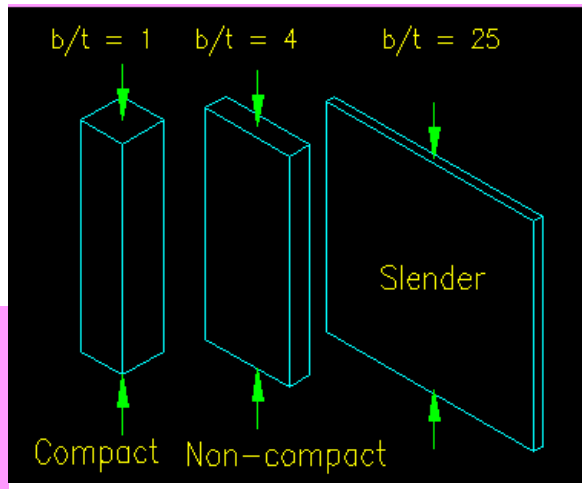


Figure 6.3.5
Unstiffened Elements
Click on image for larger view

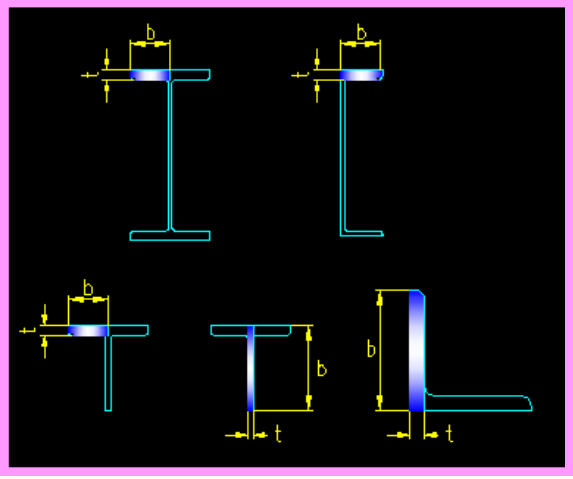
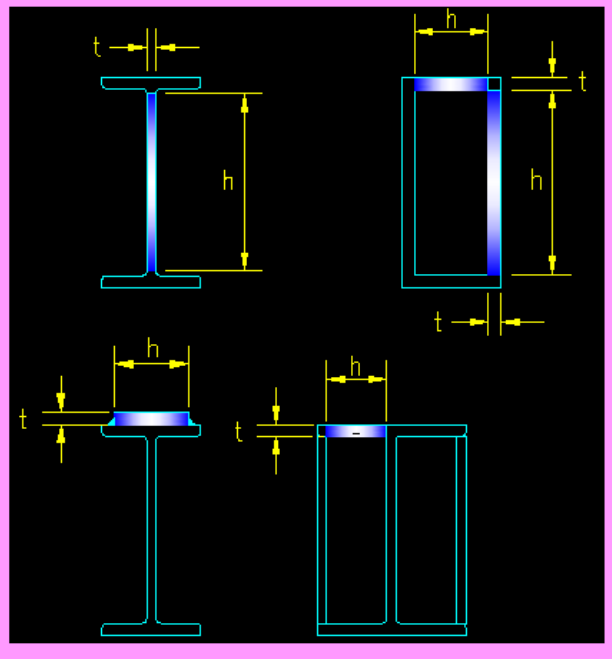
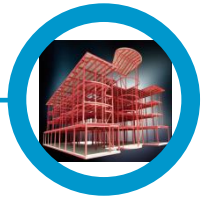


Figure 6.3.6
Stiffened Elements
Click on image for larger view





Design Example

Example 5-2

(a) Using the column critical stress values in Table 4-22 of the Manual, determine the LRFD design strength $\phi_c P_n$ and the ASD allowable strength $\frac{P_n}{\Omega_c}$ for the column shown in Fig. 5.8, if a 50-ksi steel is used.

(b) Repeat the problem, using Table 4-1 of the Manual.

(c) Calculate $\phi_c P_n$ and $\frac{P_n}{\Omega_c}$, using the equations of AISC Section E3.

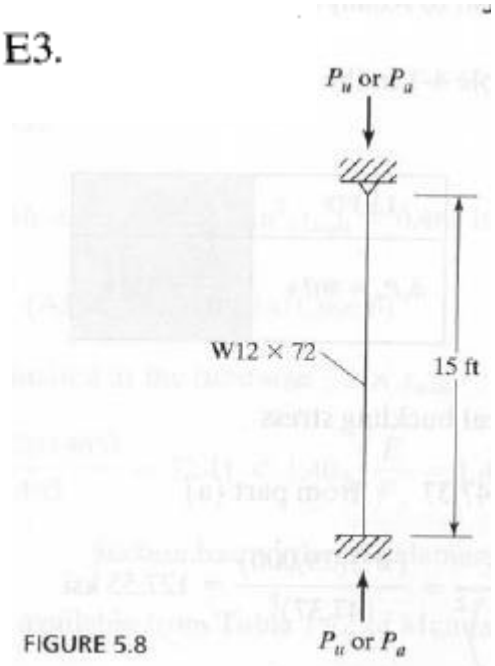
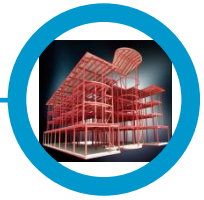


FIGURE 5.8



Design Example

Solution

(a) Using a W12 × 72 ($A = 21.1$ in^2, $r_x = 5.31$ in, $r_y = 3.04$ in, $d = 12.3$ in, $b_f = 12.00$ in, $t_f = 0.670$ in, $k = 1.27$ in, $t_w = 0.430$ in)

$$\frac{b}{t} = \frac{12.00/2}{0.670} = 8.96 < 0.56\sqrt{\frac{E}{F_y}} = 0.56\sqrt{\frac{29{,}000}{50}} = 13.49$$

∴ Nonslender unstiffened flange element

$$\frac{h}{t_w} = \frac{d - 2\,\text{k}}{t_w} = \frac{12.3 - 2(1.27)}{0.430} = 22.70 < 1.49\sqrt{\frac{E}{F_y}} = 1.49\sqrt{\frac{29{,}000}{50}} = 35.88$$

∴ Nonslender stiffened web element

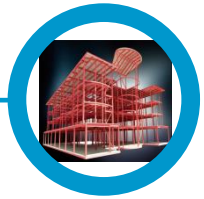
$K = 0.80$ from Table 5.1.

Obviously, $(KL/r)_y > (KL/r)_x$ and thus controls

$$\left(\frac{KL}{r}\right)_y = \frac{(0.80)(12 \times 15)\text{ in}}{3.04\text{ in}} = 47.37$$

By straight-line interpolation, $\phi_c F_{cr} = 38.19$ ksi, and $\frac{F_{cr}}{\Omega_c} = 25.43$ ksi from Table 4-22 in the Manual using $F_y = 50$ ksi steel

LRFD	ASD
$\phi_c P_n = \phi_c F_{cr} A_g = (38.19)(21.1) = 805.8$ k	$\frac{P_n}{\Omega_c} = \frac{F_{cr} A_g}{\Omega_c} = (25.43)(21.1) = 536.6$ k



Design Example

(b) Entering Table 4-1 in the Manual with KL (0.8)(15) = 12 ft

LRFD	ASD
$\phi_c P_n$ = 807 k	$\frac{P_n}{\Omega_c}$ = 537 k

(c) Elastic critical buckling stress

$\left(\frac{KL}{r}\right)_y = 47.37$ from part (a)

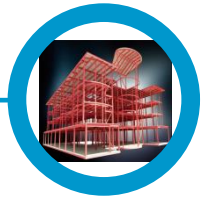
$$F_e = \frac{\pi^2 E}{\left(\frac{KL}{r}\right)^2} = \frac{(\pi^2)(29{,}000)}{(47.37)^2} = 127.55 \text{ ksi} \quad \text{(AISC Equation E3-4)}$$

Flexural buckling stress F_{cr}

$$4.71\sqrt{\frac{E}{F_y}} = 4.71\sqrt{\frac{29{,}000 \text{ ksi}}{50 \text{ ksi}}} = 113.43 > \left(\frac{KL}{r}\right)_y = 47.37$$

$$\therefore F_{cr} = \left[0.658^{\frac{F_y}{F_e}}\right]F_y = \left[0.658^{\frac{50}{127.55}}\right]50 = 42.43 \text{ ksi} \quad \text{(AISC Equation E3-2)}$$

LRFD ϕ_c = 0.90	ASD Ω_c = 1.67
$\phi_c F_{cr}$ = (0.90)(42.43) = 38.19 ksi	$\frac{F_{cr}}{\Omega_c} = \frac{42.43}{1.67}$ = 25.41 ksi
$\phi_c P_n = \phi_c F_{cr} A$ = (38.19)(21.1) = 805.8 k	$\frac{P_n}{\Omega_c} = \frac{F_{cr}}{\Omega_c} A$ = (25.41)(21.1) = 536.2 k



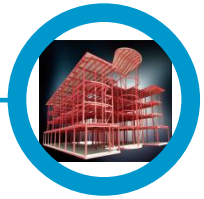
Design Example

Example 6-1

Using $F_y = 50$ ksi, select the lightest W14 available for the service column loads $P_D = 130$ k and $P_L = 210$ k. $KL = 10$ ft.

Solution

LRFD	ASD
$P_u = (1.2)(130\text{ k}) + (1.6)(210\text{ k}) = 492\text{ k}$	$P_a = 130\text{ k} + 210\text{ k} = 340\text{ k}$
Assume $\frac{KL}{r} = 50$	Assume $\frac{KL}{r} = 50$
Using $F_y = 50$ ksi steel	Using $F_y = 50$ ksi steel
$\phi_c F_{cr}$ from AISC Table 4-22 = 37.5 ksi	$\frac{F_{cr}}{\Omega_c} = 24.9$ ksi (AISC Table 4-22)
A Reqd $= \frac{P_u}{\phi_c F_{cr}} = \frac{492\text{ k}}{37.5\text{ ksi}} = 13.12\text{ in}^2$	A Reqd $= \frac{P_a}{F_{cr}/\Omega} = \frac{340\text{ k}}{24.9\text{ ksi}} = 13.65\text{ in}^2$
Try W14 × 48 ($A = 14.1\text{ in}^2$, $r_x = 5.85$ in, $r_y = 1.91$ in)	Try W14 × 48 ($A = 14.1\text{ in}^2$, $r_x = 5.85$ in, $r_y = 1.91$ in)
$\left(\frac{KL}{r}\right)_y = \frac{(12\text{ in/ft})(10\text{ ft})}{1.91\text{ in}} = 62.83$	$\left(\frac{KL}{r}\right)_y = \frac{(12\text{ in/ft})(10\text{ ft})}{1.91\text{ in}} = 62.83$
$\phi_c F_{cr} = 33.75$ ksi from AISC Table 4-22	$\frac{F_{cr}}{\Omega_c} = 22.43$ ksi from AISC Table 4-22
$\phi_c P_n = (33.75\text{ ksi})(14.1\text{ in}^2)$ $= 476\text{ k} < 492\text{ k}$ N.G.	



Design Example

Try next larger section W14 × 53 ($A = 15.6$ in^2, $r_y = 1.92$ in)

$$\left(\frac{KL}{r}\right)_y = \frac{(12 \text{ in/ft})(10 \text{ ft})}{1.92 \text{ in}} = 62.5$$

$\phi_c F_{cr} = 33.85$ ksi

$\phi_c P_n = (33.85 \text{ ksi})(15.6 \text{ in}^2)$

$= 528 \text{ k} > 492 \text{ k}$ **OK**

Use W14 × 53.

$$\frac{P_n}{\Omega_c} = (22.43 \text{ ksi})(14.1 \text{ in}^2) = 316 \text{ k} < 340 \text{ k N.G.}$$

Try next larger section W14 × 53 ($A = 15.6$ in^2, $r_y = 1.92$ in).

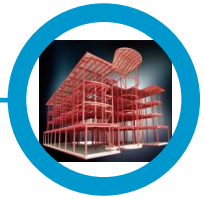
$$\left(\frac{KL}{r}\right)_y = \frac{(12 \text{ in/ft})(10 \text{ ft})}{1.92 \text{ in}} = 62.5$$

$$\frac{F_{cr}}{\Omega_c} = 22.5 \text{ ksi}$$

$$\frac{P_n}{\Omega_c} = (22.5 \text{ ksi})(15.6 \text{ in}^2) = 351 \text{ k} > 340 \text{ k OK}$$

Use W14 × 53.

Note: Table 4-1 does not indicate that a W14 × 53 is a slender member for compression.



Design Example

Table 4-1 (continued)
Available Strength in Axial Compression, kips
W-Shapes

W14

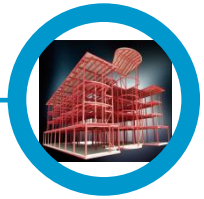
F_y = 50 ksi

Shape		W14×													
lb/ft		82		74		68		61		53		48		43[c]	
Design		P_n/Ω_c ASD	$\phi_c P_n$ LRFD	P_n/Ω_c ASD	$\phi_c P_n$ LRFD	P_n/Ω_c ASD	$\phi_c P_n$ LRFD	P_n/Ω_c ASD	$\phi_c P_n$ LRFD	P_n/Ω_c ASD	$\phi_c P_n$ LRFD	P_n/Ω_c ASD	$\phi_c P_n$ LRFD	P_n/Ω_c ASD	$\phi_c P_n$ LRFD
Effective length, KL (ft), with respect to least radius of gyration, r_y	0	719	1080	653	981	599	900	536	805	467	702	422	634	374	562
	6	676	1020	614	922	562	845	503	756	421	633	380	572	339	510
	7	661	993	600	902	550	826	492	739	406	610	366	551	327	491
	8	644	968	585	879	536	805	479	720	389	585	351	527	312	470
	9	626	940	568	854	520	782	465	699	371	557	334	502	297	447
	10	606	910	550	827	503	756	450	676	351	528	316	475	281	422
	11	584	878	531	797	485	729	433	651	331	497	298	447	264	397
	12	562	844	510	767	466	701	416	626	310	465	279	419	247	371
	13	538	809	489	735	446	671	398	599	288	433	259	390	229	345
	14	514	772	467	701	426	640	380	571	267	401	240	360	212	318
	15	489	735	444	667	405	608	361	543	246	369	221	331	194	292
	16	464	697	421	633	384	577	342	514	225	338	202	303	177	267
	17	438	659	398	598	362	544	323	485	205	308	183	276	161	242
	18	413	620	375	563	341	512	304	456	185	278	166	249	145	218
	19	387	582	352	529	320	480	285	428	166	250	149	224	130	196
	20	362	545	329	495	299	449	266	399	150	226	134	202	117	177
	22	314	472	285	428	258	388	229	345	124	186	111	167	97.1	146
	24	267	402	243	365	219	330	195	293	104	157	93.2	140	81.6	123
	26	228	343	207	311	187	281	166	249	88.8	133	79.4	119	69.5	104
	28	197	295	179	268	161	242	143	215	76.6	115	68.5	103	59.9	90.1
	30	171	257	156	234	140	211	125	187	66.7	100	59.7	89.7	52.2	78.5
	32	150	226	137	205	123	185	110	165	58.6	88.1				
	34	133	200	121	182	109	164	97.0	146						
	36	119	179	108	162	97.5	147	86.5	130						
	38	107	160	96.9	146	87.5	131	77.7	117						
	40	96.3	145	87.5	131	79.0	119	70.1	105						

Properties															
P_{wo}, kips	123	185	104	155	90.6	136	77.5	116	77.1	116	67.4	101	56.9	85.4
P_{wl}, kips/in.	17.0	25.5	15.0	22.5	13.8	20.8	12.5	18.8	12.3	18.5	11.3	17.0	10.2	15.3
P_{wb}, kips	201	302	138	207	108	163	80.1	120	76.7	115	59.5	89.5	43.0	64.7
P_{fb}, kips	137	206	115	173	97.0	146	77.8	117	81.5	123	66.2	99.6	52.6	79.0
L_p, ft	8.76		8.76		8.69		8.65		6.78		6.75		6.68	
L_r, ft	33.2		31.0		29.3		27.5		22.3		21.1		20.0	
A_g, in.^{2}	24.0		21.8		20.0		17.9		15.6		14.1		12.6	
I_x, in.^{4}	881		795		722		640		541		484		428	
I_y, in.^{4}	148		134		121		107		57.7		51.4		45.2	
r_y, in.	2.48		2.48		2.46		2.45		1.92		1.91		1.89	
r_x/r_y	2.44		2.44		2.44		2.44		3.07		3.06		3.08	
$P_{ex}(KL)^2/10^4$, k-in.^{2}	25200		22800		20700		18300		15500		13900		12300	
$P_{ey}(KL)^2/10^4$, k-in.^{2}	4240		3840		3460		3060		1650		1470		1290	

ASD	LRFD
Ω_c = 1.67	ϕ_c = 0.90

[c] Shape is slender for compression with F_y = 50 ksi.
Note: Heavy line indicates KL/r_y equal to or greater than 200.



Design via AISC Design Tables

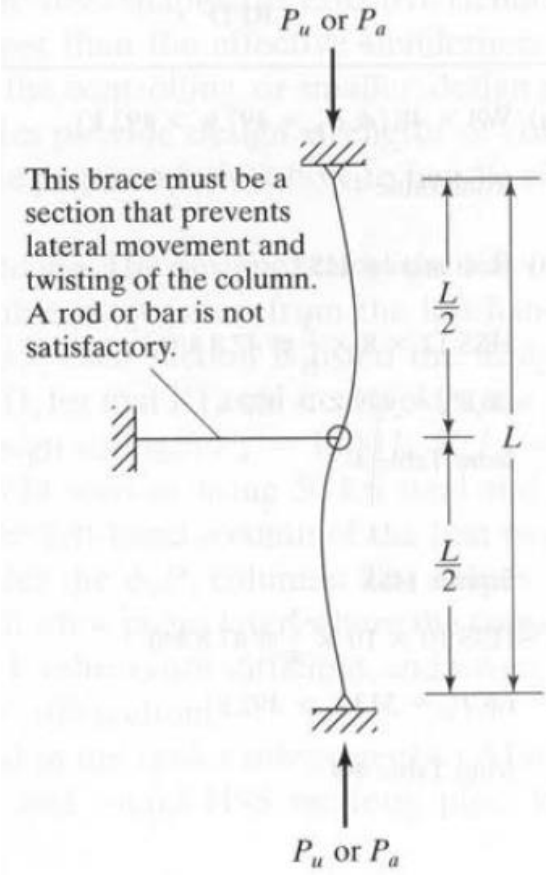
For this discussion, it is assumed that K is the same in both directions. Then, if we are to have equal strengths about the x and y axes, the following relation must hold:

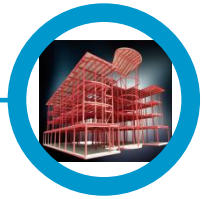
$$\frac{L_x}{r_x} = \frac{L_y}{r_y}$$

For L_y to be equivalent to L_x we would have

$$L_x = L_y \frac{r_x}{r_y}$$

If $L_y(r_x/r_y)$ is less than L_x, then L_x controls; if greater than L_x, then L_y controls.





Design Example

Example 6-3

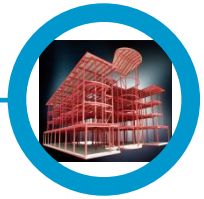
Select the lightest available W12 section, using both the LRFD and ASD methods for the following conditions: $F_y = 50$ ksi, $P_D = 250$ k, $P_L = 400$ k, $K_xL_x = 26$ ft and $K_yL_y = 13$ ft.

(a) By trial and error
(b) Using AISC tables

Solution

(a) Using trial and error to select a section, using the LRFD expressions, and then checking the section with both the LRFD and ASD methods

LRFD	ASD
$P_u = (1.2)(250\text{ k}) + (1.6)(400\text{ k}) = 940$ k	$P = 250\text{ k} + 400\text{ k} = 650$ k
Assume $\frac{KL}{r} = 50$	Assume $\frac{KL}{r} = 50$
Using $F_y = 50$ ksi steel $\phi_c F_{cr} = 37.5$ ksi (AISC Table 4-22)	Using $F_y = 50$ ksi steel $\frac{F_{cr}}{\Omega_c} = 24.9$ ksi (AISC Table 4-22)
A Reqd $= \frac{940\text{ k}}{37.5\text{ ksi}} = 25.07\text{ in}^2$	A Reqd $= \frac{650\text{ k}}{24.9\text{ ksi}} = 26.10\text{ in}^2$



Design Example

Try W12 × 87 ($A = 25.6\ \text{in}^2$, $r_x = 5.38$ in, $r_y = 3.07$ in)

$$\left(\frac{KL}{r}\right)_x = \frac{(12\ \text{in/ft})(26\ \text{ft})}{5.38\ \text{in}} = 57.99 \leftarrow \therefore \left(\frac{KL}{r}\right)_x \text{ controls}$$

$$\left(\frac{KL}{r}\right)_y = \frac{(12\ \text{in/ft})(13\ \text{ft})}{3.07\ \text{in}} = 50.81$$

$\phi_c F_{cr} = 35.2$ ksi (Table 4-22)

$\phi_c P_n = (35.2\ \text{ksi})(25.6\ \text{in}^2)$

$= 901\ \text{k} < 940\ \text{k}$ **N.G.**

Try W12 × 87 ($A = 25.6\ \text{in}^2$, $r_x = 5.38$ in, $r_y = 3.07$ in)

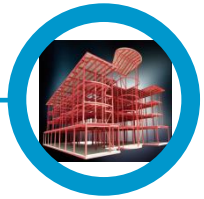
$$\left(\frac{KL}{r}\right)_x = \frac{(12\ \text{in/ft})(26\ \text{ft})}{5.38\ \text{in}} = 57.99 \leftarrow \therefore \left(\frac{KL}{r}\right)_x \text{ controls}$$

$$\left(\frac{KL}{r}\right)_y = \frac{(12\ \text{in/ft})(13\ \text{ft})}{3.07\ \text{in}} = 50.81$$

$\dfrac{F_{cr}}{\Omega_c} = 23.4$ ksi (Table 4-22)

$\dfrac{P_n}{\Omega_c} = (23.4\ \text{ksi})(25.6\ \text{in}^2)$

$= 599\ \text{k} < 650\ \text{k}$ **N.G.**



Design Example

A subsequent check of the next-larger W12 section, a W12 × 96, shows that it will work for both the LRFD and ASD procedures.

(b) Using AISC Tables. Assuming K_yL_y controls

Enter Table 4-1 with $K_yL_y = 13$ ft, $F_y = 50$ ksi and $P_u = 940$ k

LRFD

Try W12 × 87 $\left(\frac{r_x}{r_y} = 1.75\right)$; $\phi P_n = 954$ k

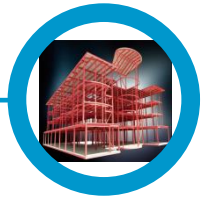
Equivalent $K_yL_y = \dfrac{K_xL_x}{\frac{r_x}{r_y}}$

$= \dfrac{26}{1.75} = 14.86 > K_yL_y$ of 13 ft. $\therefore$ K_xL_x controls

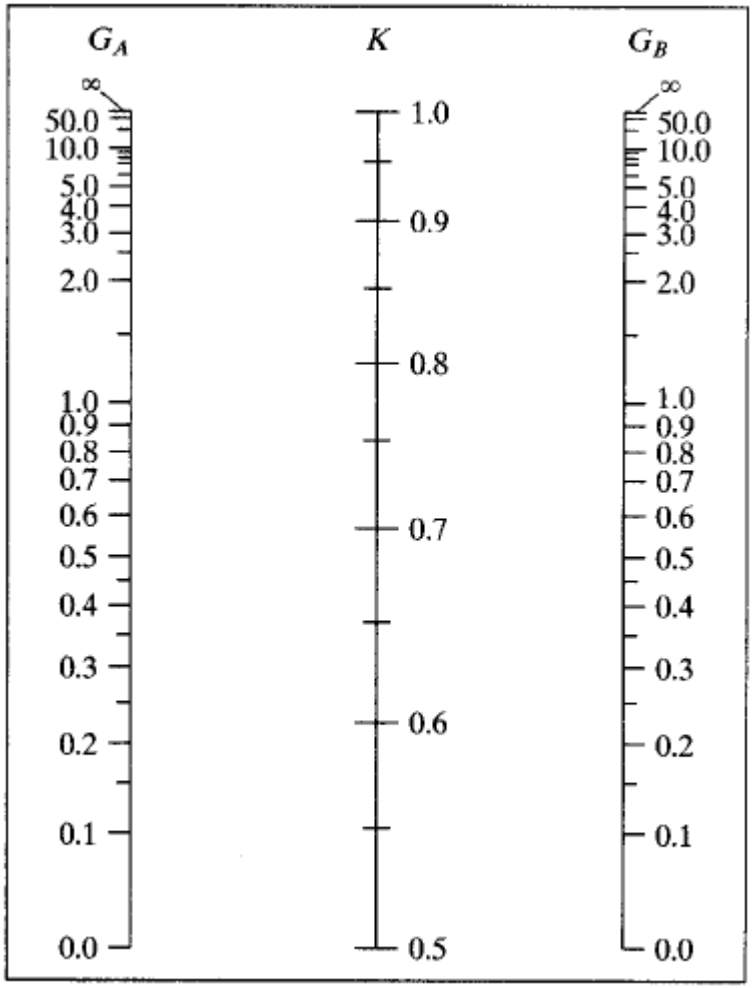
Use $K_yL_y = 14.86$ ft and reenter tables

LRFD	ASD
Use W12 × 96	**Use W12 × 96**
$\phi_c P_n = 994$ k > 940 k **OK**	$\frac{P_n}{\Omega_c} = 662$ k > 650 k **OK**

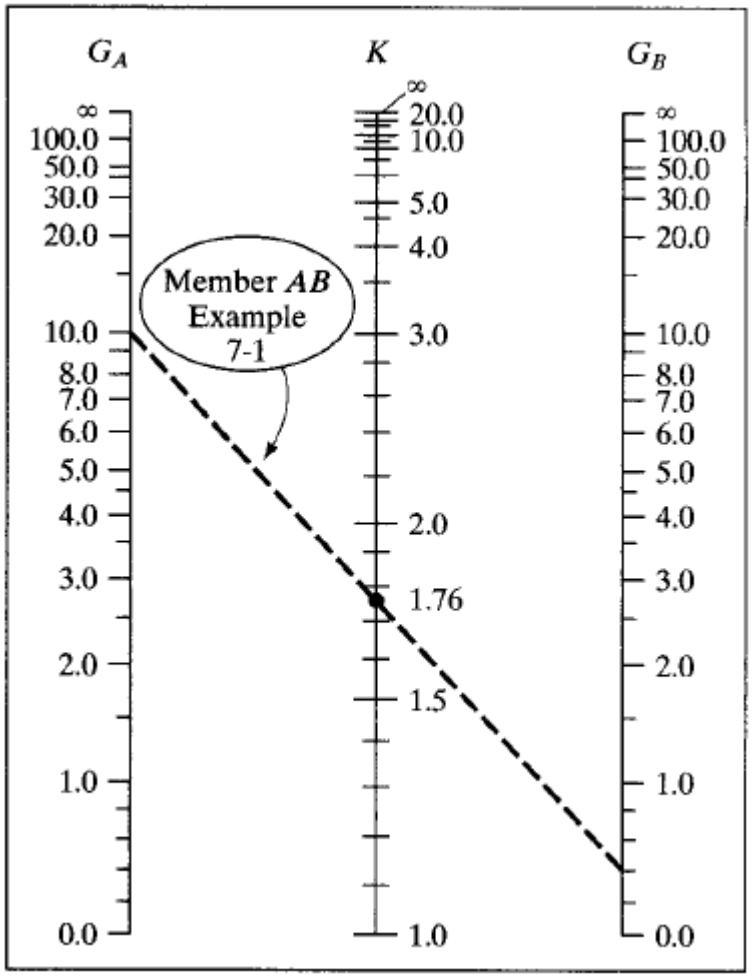
Note: Table 4-1 does not indicate that the W12 × 96 is a slender member for compression.



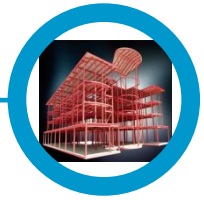
Effective length via charts



(a) Sidesway inhibited (Braced Frame)



(b) Sidesway uninhibited (Moment Frame)



Effective length via charts

$$G = \frac{\sum\left(\frac{E_c I_c}{L_c}\right)}{\sum\left(\frac{E_g I_g}{L_g}\right)}$$

AISC Equation (C-A-7-2)

The symbol Σ indicates a summation of all members rigidly connected to that joint and located in the plane in which buckling of the column is being considered. E_c is the elastic modulus of the column, I_c is the moment of inertia of the column, and L_c is the unsupported length of the column. E_g is the elastic modulus of the girder, I_g is the moment of inertia of the girder, and L_g is the unsupported length of the girder or other restraining member. I_c and I_g are taken about axes perpendicular to the plane of buckling being considered. The alignment charts are valid for different materials if an appropriate effective rigidity, EI, is used in the calculation of G.

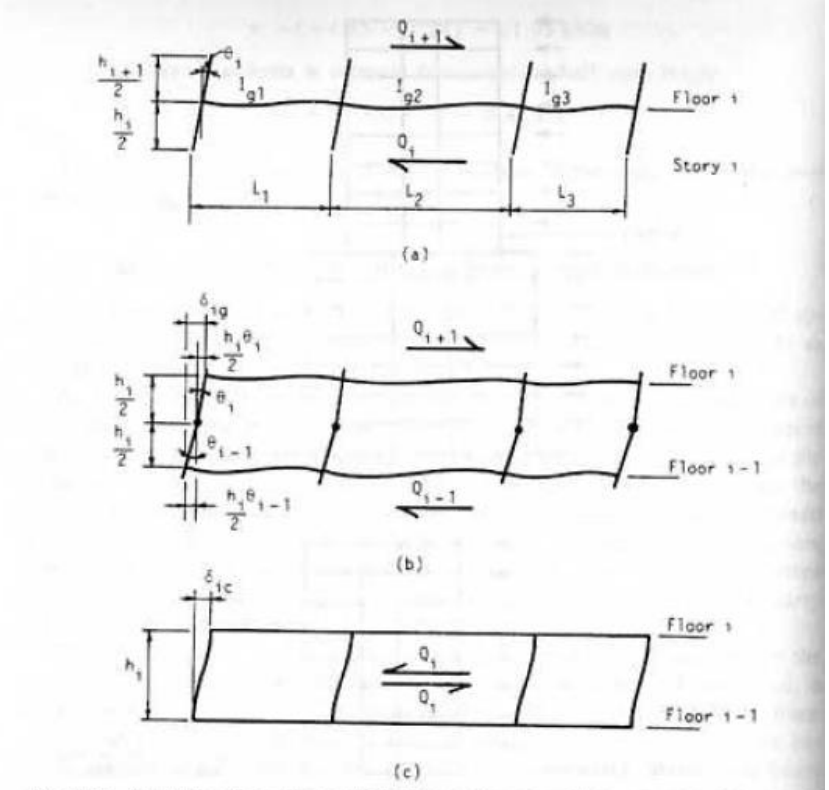
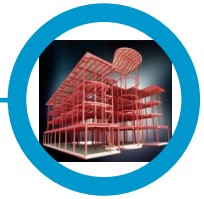


Fig. 7.13 (a) Joint rotation due to girder flexure; (b) story drift due to girder flexure; (c) story drift due to column flexure.

Story Drift due to Girder Flexure.

Story Drift due to Column Flexure.

Story Drift due to Overall Bending.



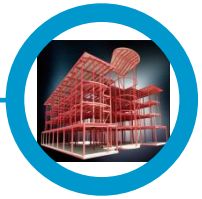
Effective length via charts

To determine value of K factors for columns of steel frame by alignment charts, the following steps are taken:

- Select the appropriate chart (sidesway inhibited or sidesway uninhibited)
- Compute G at each end of the column and label the values G_A and G_B as desired

In applying the charts, the *G* factors at the column bases are quite variable. It is recommended that the following two rules be applied to obtain their values:

1. For pinned columns, *G* is theoretically infinite, such as when a column is connected to a footing with a frictionless hinge. Since such a connection is not frictionless, it is recommended that *G* be made equal to 10 where such nonrigid supports are used.
2. For rigid connections of columns to footings, *G* theoretically approaches zero, but from a practical standpoint, a value of 1.0 is recommended, because no connections are perfectly rigid.



Effective length via charts

Example 7-1

Determine the effective length factor for each of the columns of the frame shown in Fig. 7.4 if the frame is not braced against sidesway. Use the alignment charts of Fig. 7.2(b).

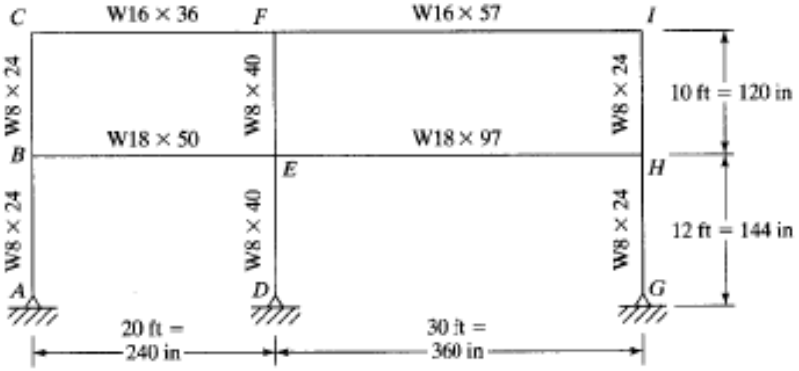
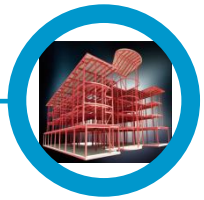


FIGURE 7.4

Solution. Stiffness factors: E is assumed to be 29,000 ksi for all members and is therefore neglected in the equation to calculate G.

	Member	Shape	I	L	I/L
Columns	AB	W8 × 24	82.7	144	0.574
	BC	W8 × 24	82.7	120	0.689
	DE	W8 × 40	146	144	1.014
	EF	W8 × 40	146	120	1.217
	GH	W8 × 24	82.7	144	0.574
	HI	W8 × 24	82.7	120	0.689
Girders	BE	W18 × 50	800	240	3.333
	CF	W16 × 36	448	240	1.867
	EH	W18 × 97	1750	360	4.861
	FI	W16 × 57	758	360	2.106



Effective length via charts

G factors for each joint:

Joint	$\Sigma(I_c/L_c)/\Sigma(I_g/L_g)$	G
A	Pinned Column, $G = 10$	10.0
B	$\frac{0.574 + 0.689}{3.333}$	0.379
C	$\frac{0.689}{1.867}$	0.369
D	Pinned Column, $G = 10$	10.0
E	$\frac{1.014 + 1.217}{(3.333 + 4.861)}$	0.272
F	$\frac{1.217}{(1.867 + 2.106)}$	0.306
G	Pinned Column, $G = 10$	10.0
H	$\frac{0.574 + 0.689}{4.861}$	0.260
I	$\frac{0.689}{2.106}$	0.327

Column K factors from chart [Fig. 7.2(b)]:

Column	G_A	G_B	K*
AB	10.0	0.379	1.76
BC	0.379	0.369	1.12
DE	10.0	0.272	1.74
EF	0.272	0.306	1.10
GH	10.0	0.260	1.73
HI	0.260	0.327	1.10

*It is a little difficult to read the charts to the three decimal places shown by the author. He has used a larger copy of Fig. 7.2 for his work. For all practical design purposes, the K values can be read to two places, which can easily be accomplished with this figure.


Questions?